

(54) **APPARARATUS, METHOD AND ARTICLE TO FACILITATE AUTOMATIC DETECTION AND REMOVAL OF FRAUDULENT USER INFORMATION IN A NETWORK ENVIRONMENT**

(71) Applicant: **PLENTYOFFISH MEDIA ULC,**
Vancouver (CA)

(72) Inventors: **Thomas Levi,** Vancouver (CA); **Steve Oldridge,** Vancouver (CA)

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(60) Provisional application No. 61/911,908, filed on Dec. 4, 2013.

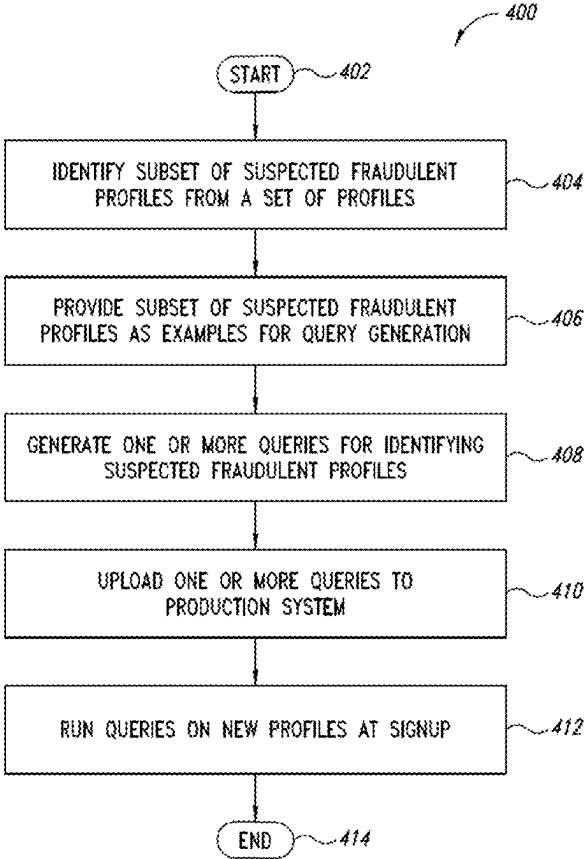
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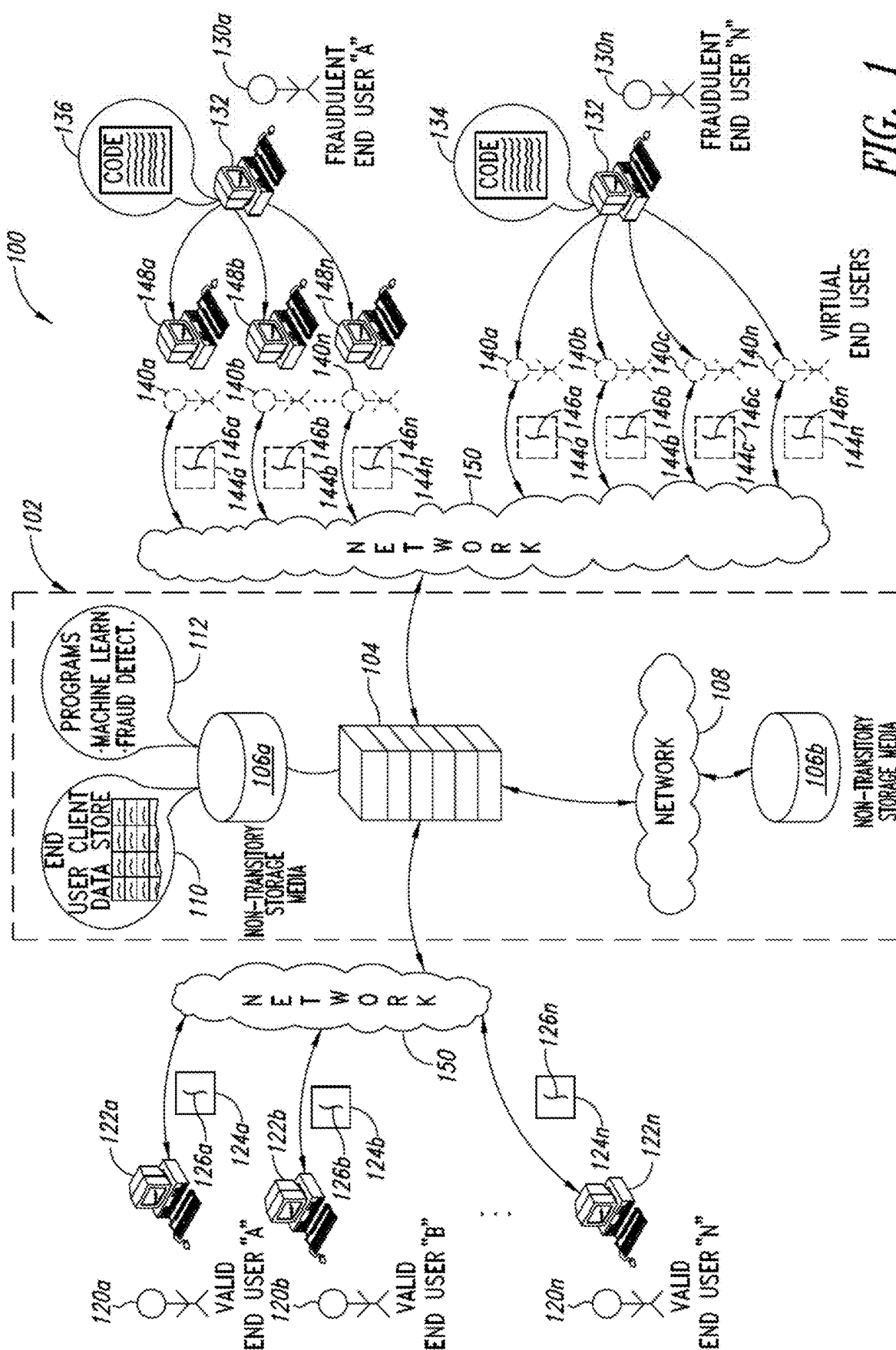
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(57) **ABSTRACT**

A fraud detection system obtains a number of known fraudulent end-user profiles and/or otherwise undesirable end-user profiles. Using statistical analysis techniques that include clustering the end-user profiles by attributes and attribute values and/or combinations of attributes and attribute values, the fraud detection system identifies on a continuous, periodic, or aperiodic basis those attribute values and/or attribute value combinations that appear in fraudulent or otherwise undesirable end-user profiles. Using this data, the fraud detection system generates one or more queries to identify those end-user profiles having attribute values or combinations of attribute values that likely indicate a fraudulent or otherwise undesirable end-user profile. The fraud detection system runs these queries against incoming registrations to identify and screen fraudulent end-user profiles from entering the system and also runs these queries against stored end-user profile databases to identify and remove fraudulent or otherwise undesirable end-user profiles from the end-user database.





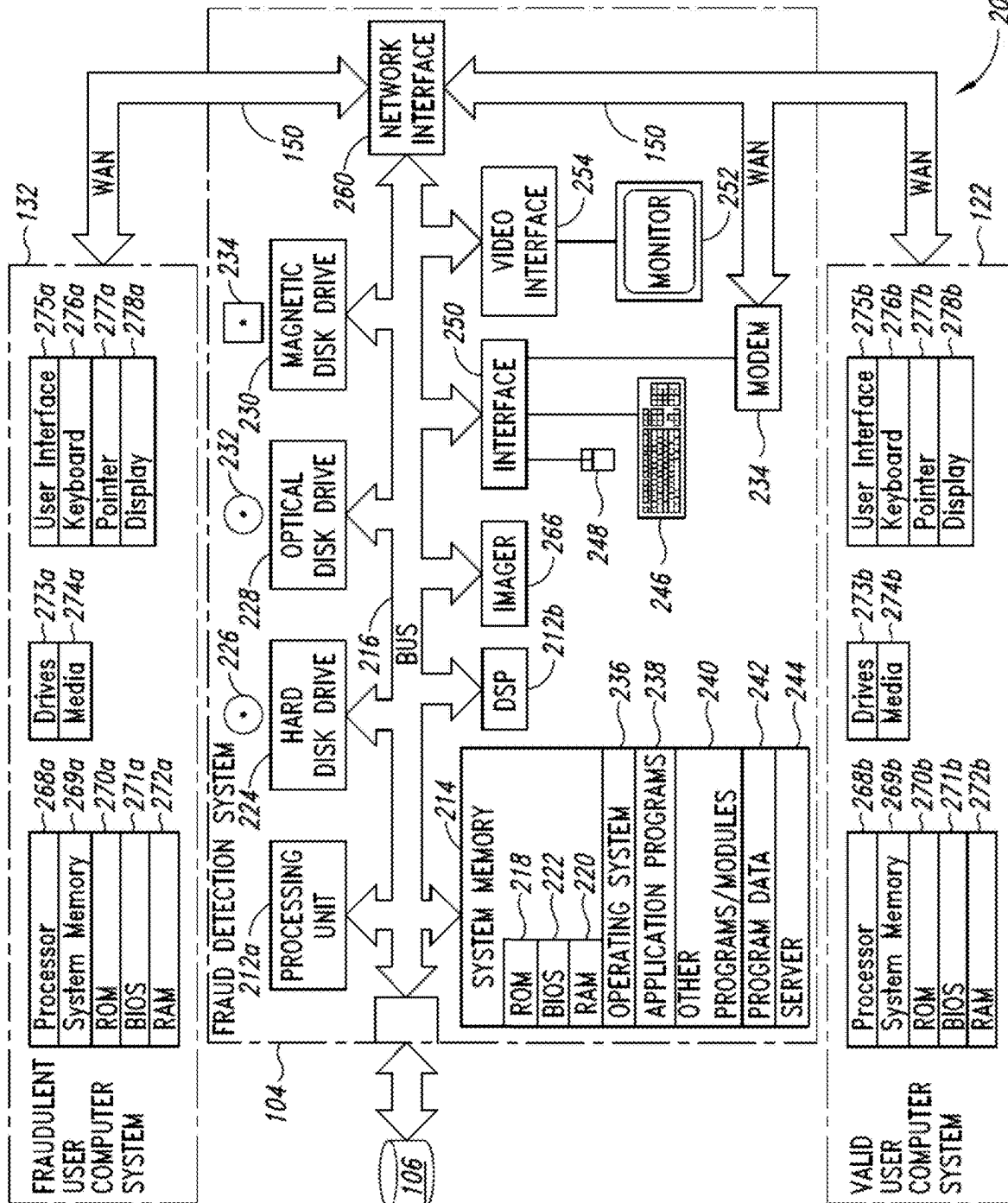


FIG. 2

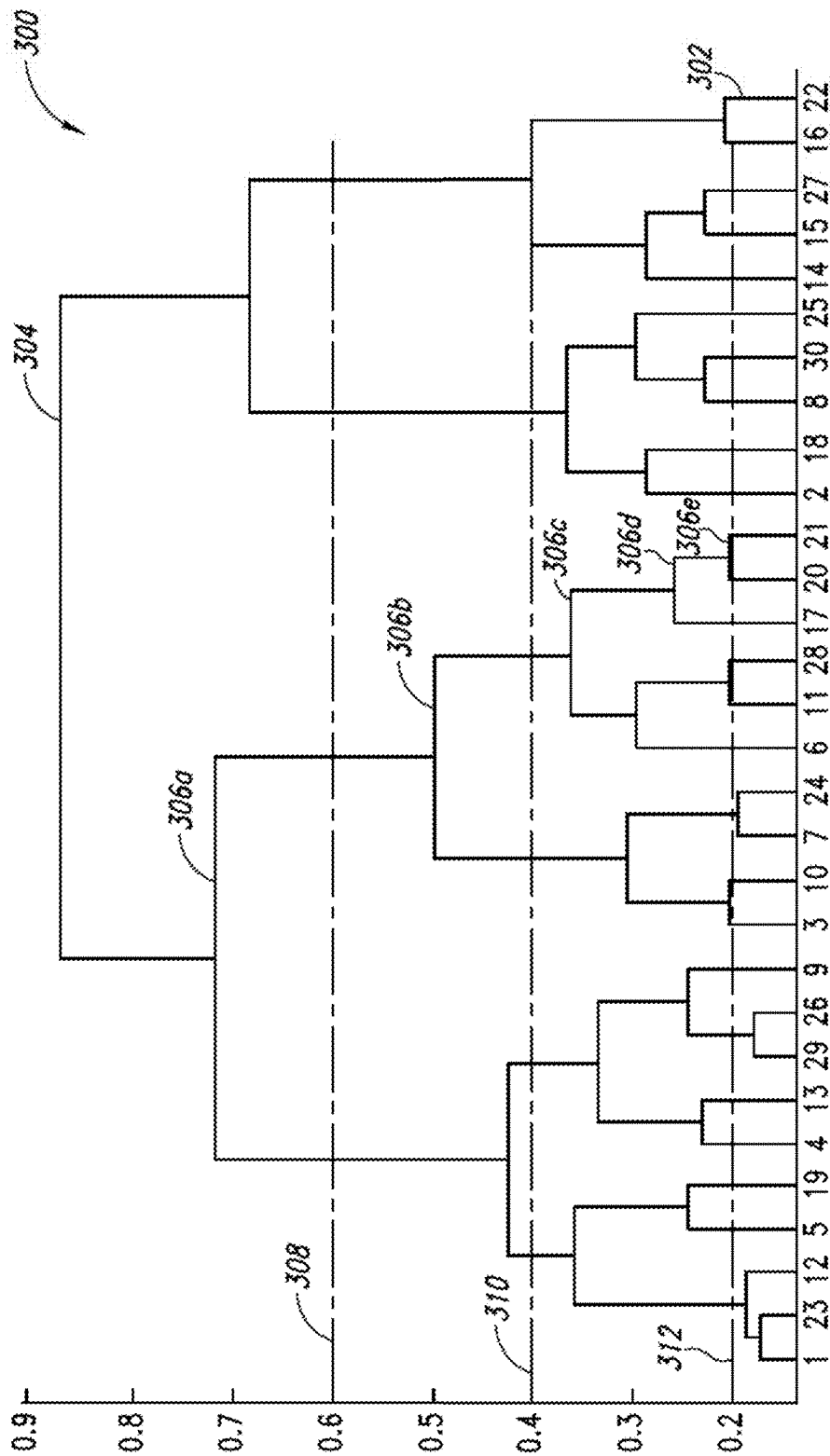
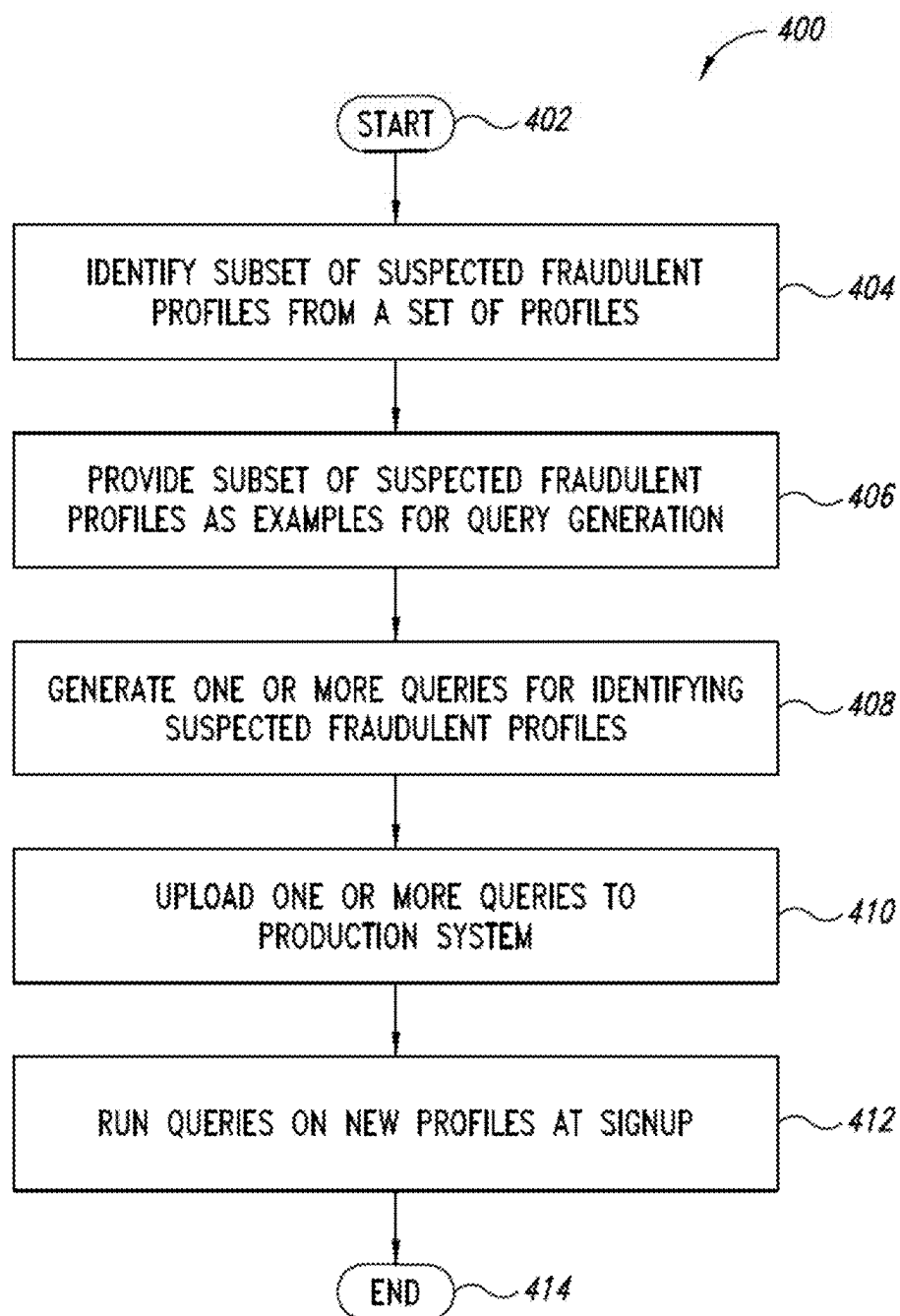
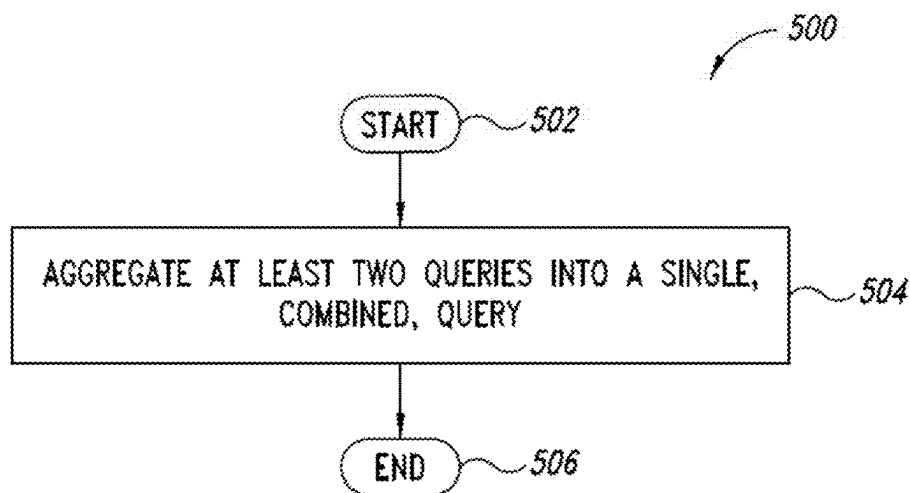
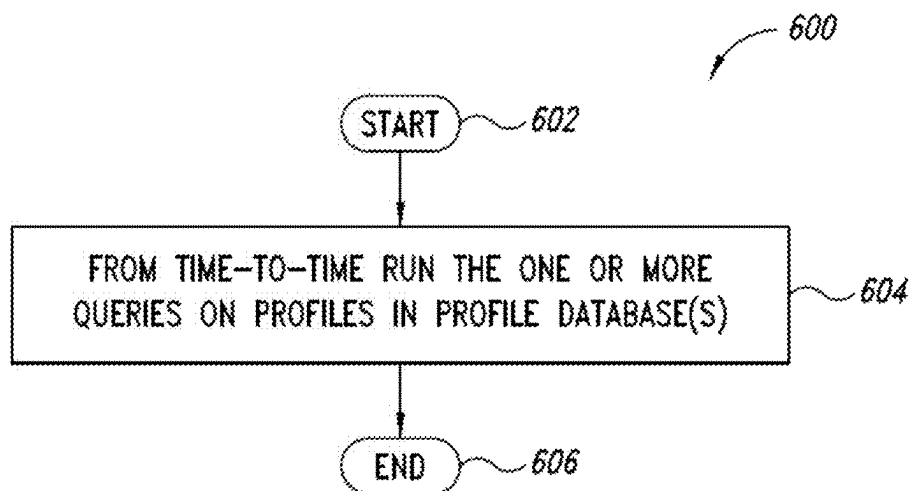


FIG. 3

*FIG. 4*

*FIG. 5**FIG. 6*

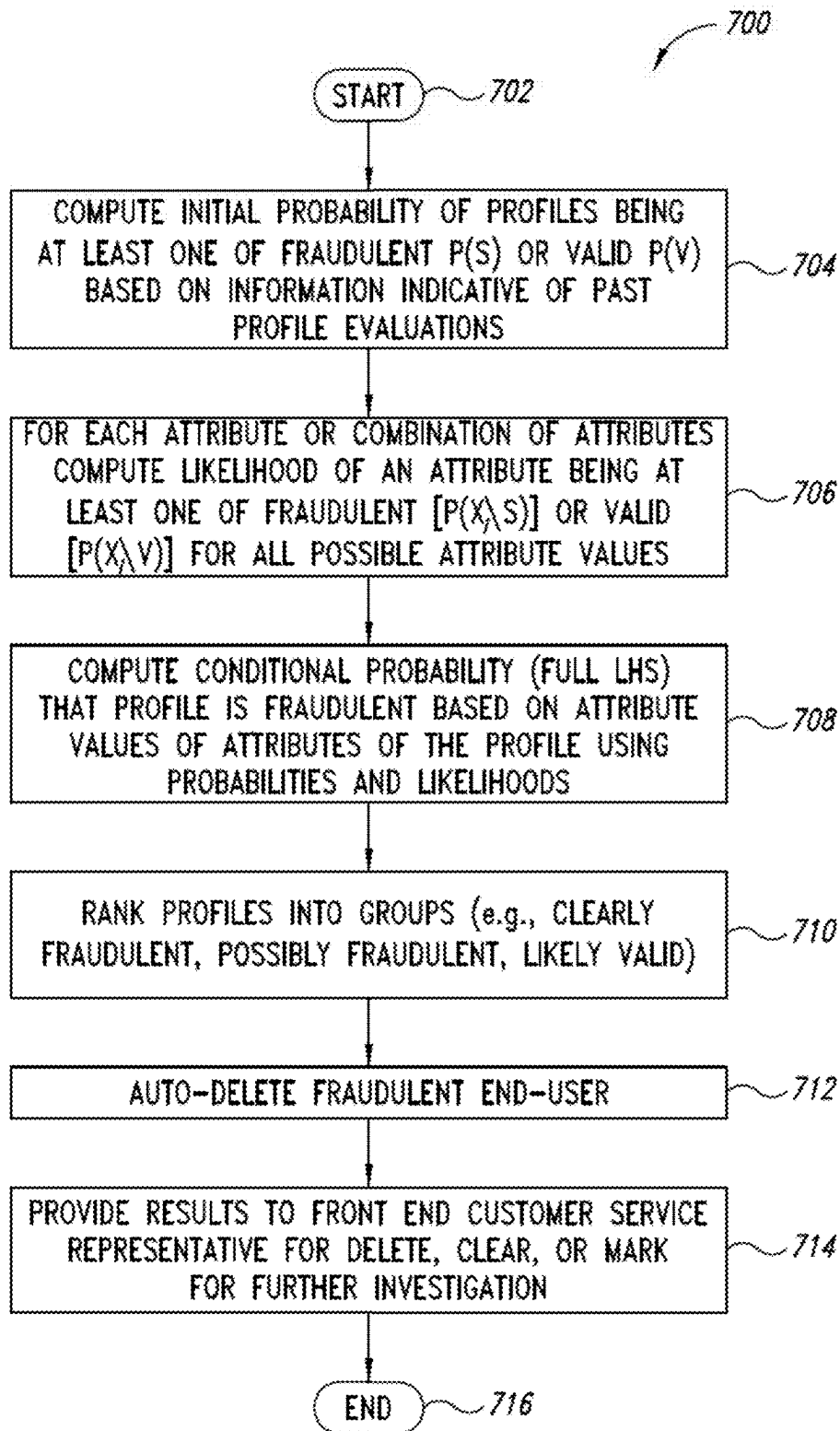
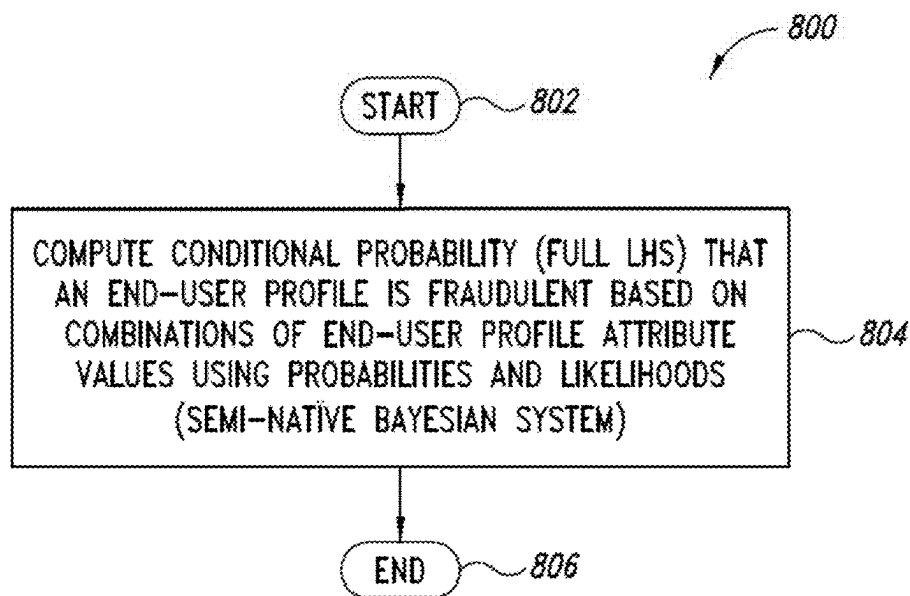
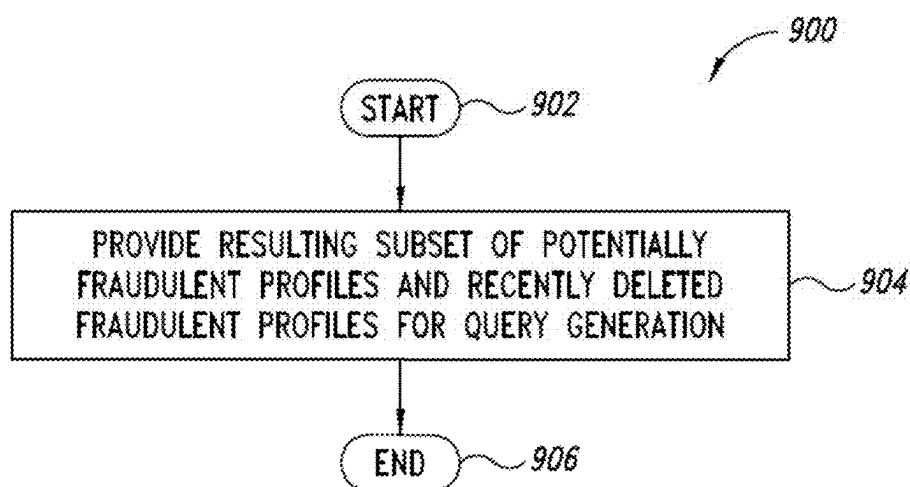


FIG. 7

*FIG. 8**FIG. 9*

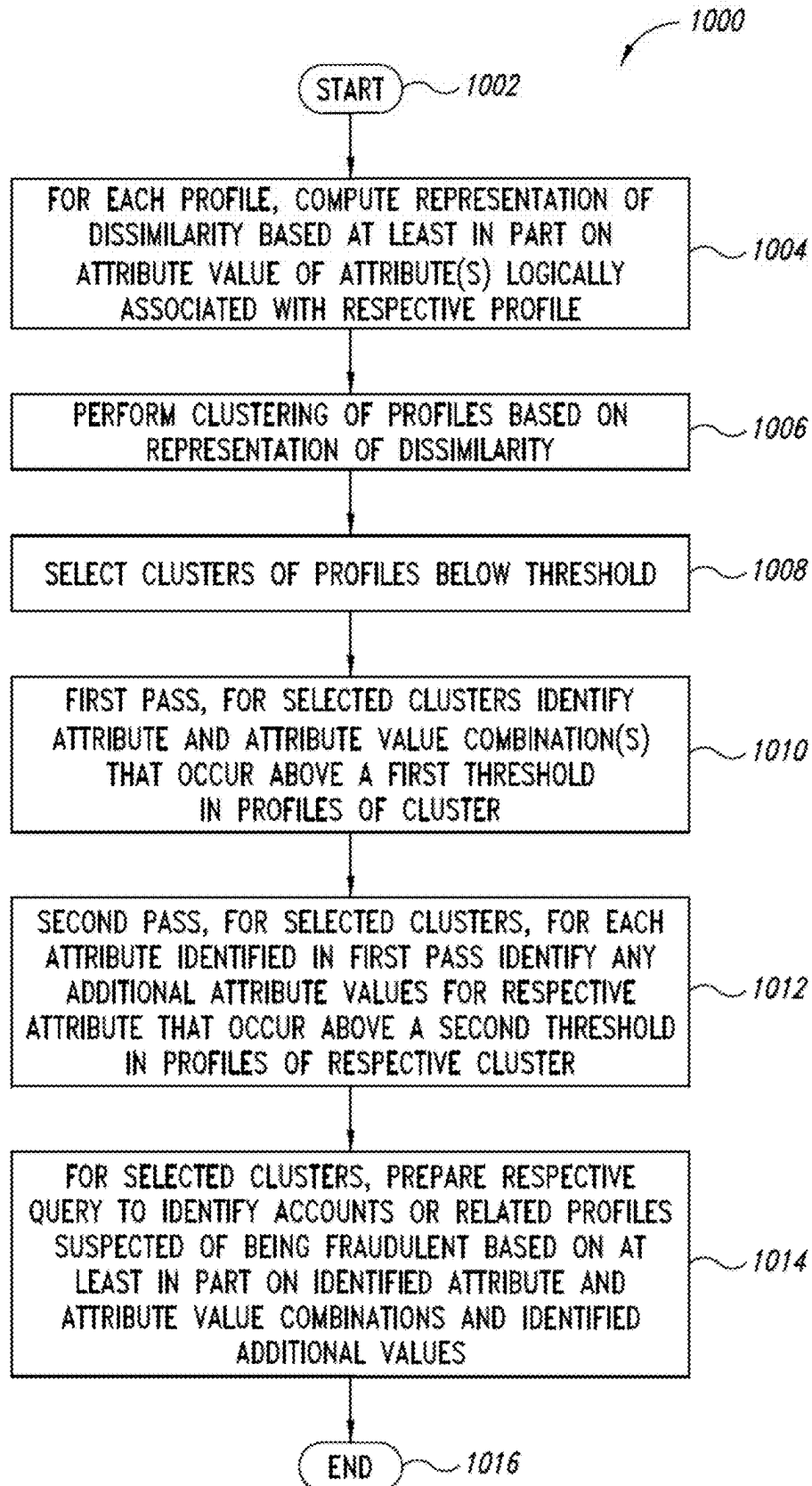
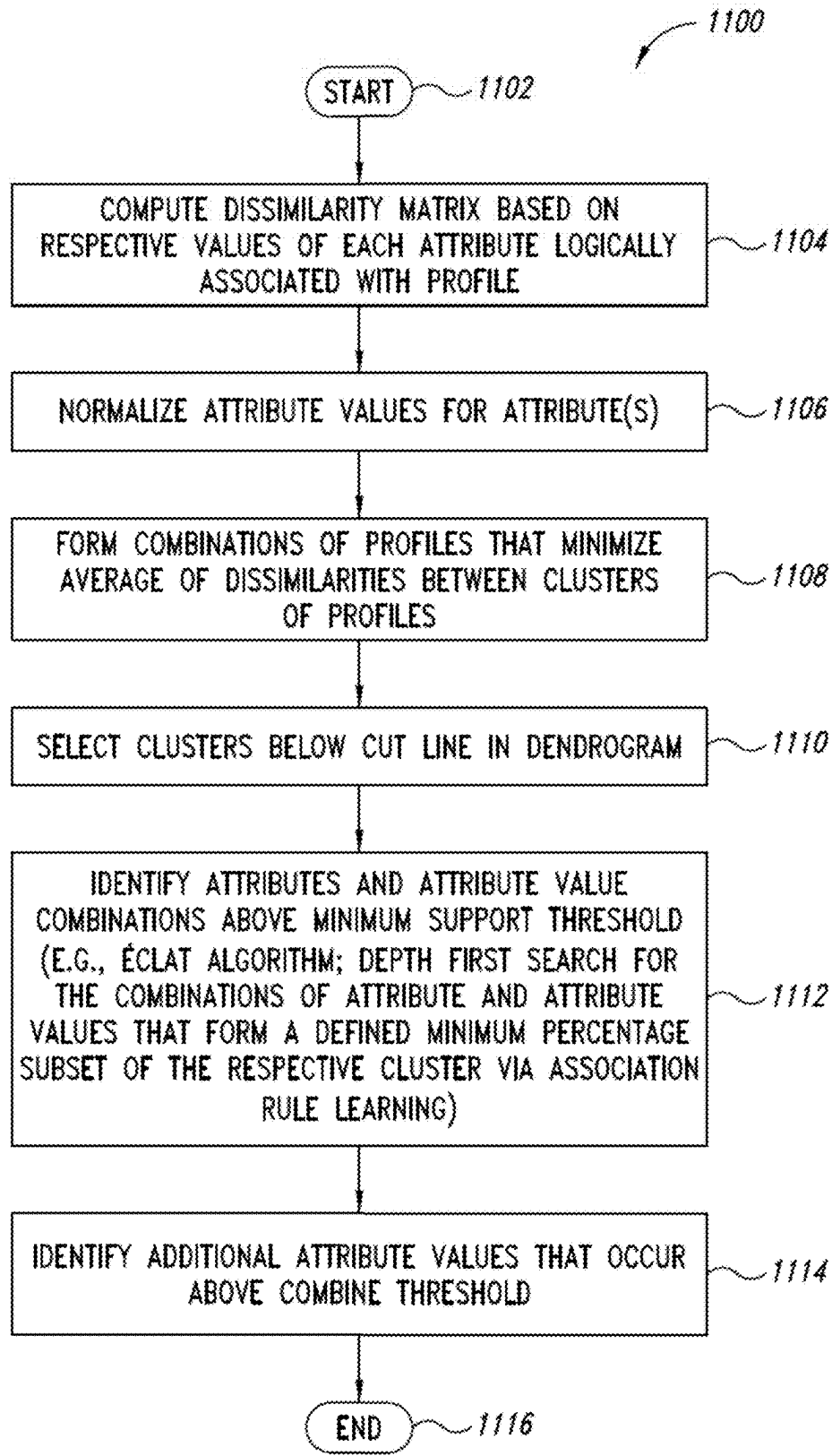


FIG. 10

*FIG. 11*

**APPARATUS, METHOD AND ARTICLE
TO FACILITATE AUTOMATIC DETECTION
AND REMOVAL OF FRAUDULENT USER
INFORMATION IN A NETWORK
ENVIRONMENT**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is a continuation of U.S. patent application Ser. No. 18/622,553, filed Mar. 29, 2024, which is a continuation of U.S. patent application Ser. No. 18/069,866, filed Dec. 21, 2022, now U.S. Pat. No. 11,949,747, issued Apr. 2, 2024, which is a continuation of U.S. patent application Ser. No. 16/833,427, filed Mar. 27, 2020, now U.S. Pat. No. 11,546,433, issued Jan. 3, 2023, which is a continuation of U.S. patent application Ser. No. 15/782,576, filed Oct. 12, 2017, now U.S. Pat. No. 10,637,959, issued Apr. 28, 2020, which is a continuation of U.S. patent application Ser. No. 14/561,004, filed Dec. 4, 2014, now U.S. Pat. No. 9,870,465, issued Jan. 16, 2018, which claims priority to Provisional Application Ser. No. 61/911,908, filed Dec. 4, 2013, which are hereby incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure generally relates to computing systems and methods, and in particular to systems and methods that facilitate the detection of suspected fraudulent user profiles in computing systems and networks.

Description of the Related Art

[0003] The term “Web 2.0” has been used to describe Web sites providing information and entertainment to users in a non-static format. In particular Web 2.0 has often been used to describe content delivered using a person-to-person or person-to-group environment such as those found on social networking sites, blogs, wikis, folksonomies, and video sharing sites. In many instances, users of such sites may provide personal or other information either as part of a site registration process, or in dialog with other site users. In many instances, such information is used to facilitate linking between site user who may share similar interests, personalities, or profiles. In some instances, users who are “matched” or otherwise introduced to each other by the Website may exchange additional information either as a “private” or similar connection between users through the Website itself or external to the Website, for example via direct or screened email exchanges between users. Users may exchange information through relationships developed with Website users, including information of a personal nature such as photographs, financial information, family information, and the like either through the Website or external to the Website (e.g., via email).

[0004] While many users are scrupulous in their dealings with other users in such environments, on occasion less scrupulous users may also participate in an attempt to extract information of a personal nature from unwary users. Among other things, such information may be used to commit identity theft, blackmail, fraud, phishing, misdirection, click fraud, or for other nefarious purposes. For example, in some instances, a scammer may generate a number of false or

shadow profiles to interact with other users and, on occasion extract personal information from those users. At times, tens, hundreds, or even thousands of false or shadow users may be generated through the use of a software routine that repeatedly connects to a Website and creates a user profile. In some instances, a plurality of computers may be infected with malware and operated covertly as “bots” in a bot network. For example, a plurality of computers operating as bots may be autonomously or manually controlled or caused to generate fraudulent user profiles. The computers or bots typically use a random or semi-random mixture of registration data retrieved from a defined set of user registration data selected by the scammer. Over time, left unchecked such bots can create tens, hundreds, or even thousands of false or shadow user profiles. For example, a scammer may determine that middle age men are more susceptible to exchanging information with 25-35 year old females. With such information, the scammer can create a bot to generate user profiles representative of typical 25-35 year old women. Once an unwary user begins communicating with one of the shadow users, the scammer can direct the information exchange between the shadow user and the valid site user to extract desired information from the valid site user.

BRIEF SUMMARY

[0005] Matching service entities bring together end-users identified as having one or more common interests, aspirations, and/or goals. Matching service entities include commercial matching services that connect end-users such as product or service suppliers with other end-users such as product or service consumers. Matching service entities include personal matching services that connect end-user individuals with other end-user individuals who have expressed interests in a common subject, theme, or cause. Matching service entities may also include Websites providing one or more social media services for developing, establishing, or forming short-term relationships between end-users of the Website. Yet other matching service entities include relationship matching services that connect end-users seeking a longer term relationship with other individual end-users who may also be seeking a longer term relationship.

[0006] Typically, such matching service entities directly or indirectly collect information from an end-user to establish a user profile that identifies not only the individual but also the individual’s likes, dislikes, and/or preferences. Each end-user is thus defined by an end-user profile that includes any number of logically associated end-user attributes, each having one or more associated attribute values. At least some of an end-user’s attribute values may be collected directly by the matching service entity, for example through the use of surveys and questionnaires that the end-user completes as part of the registration process to participate or sign-up with the matching service entity. At least some of an end-user’s attribute values may be collected indirectly by the matching service entity and may be collected autonomously by the matching service provider. For example, the Internet Protocol or IP address used to access the matching service entity by a particular end-user may be stored as an attribute value in the respective end-user’s profile. In another example, an end-user’s browser history may be stored as an attribute value indicative of the general interests of the respective end-user.

[0007] The matching service entity uses any number of algorithms to determine the existence of compatible matches or potential matches between end-users and to select appropriate potential “matches” for each end-user. Typically, these potential matches are presented to the end-user to follow-up via electronic message (e.g., email), telephone, or in person meetings to determine the overall level of compatibility between the end-users. In fact, much of the perceived value of such matching service entities lies in their reputation for recommending “good” or “compatible” matches for end-users. Success in the relationships formed between end-users builds the reputation of the matching service entity, increasing usage and often increasing membership and/or advertising revenue. Conversely, a matching service entity that gains a reputation for recommending “poor” or “incompatible” matches for end-users will see a loss of members and loss of membership and advertising revenue. Matching service providers therefore have an interest in recommending the best possible matches for their end-users and in ensuring the most positive experience results for each registered end-user. Unfortunately, fraudulent end-users (e.g., scammers) may register with a matching service entity to prey on unwary end-users or to otherwise frustrate the end-users and/or matching service entity. Typically, such fraudulent end-users will generate a number of fraudulent end-user profiles on the matching service entity to increase the likelihood that one of the fraudulent end-user profiles will be matched to and develop a relationship with one or more valid end-users of the matching service. Once a relationship develops between the fraudulent end-user and the valid end-user, the fraudulent end-user is often able to leverage the relationship to obtain cash and other valuables from the valid end-user, compromise the valid end-user’s computer system to perform other illegal acts, and the like. To generate a large number of fraudulent end-user profiles, the fraudulent end-user will frequently employ software, scripts, or other machine-readable instruction sets on one or more computer systems (e.g., “bots”) to autonomously generate a number of profiles having attribute values selected from one or more defined sets of attribute values. For example, a fraudulent end-user may define sets of attribute values representative of a 25-35 year-old female to generate a number of user profiles having different attribute value permutations, but all of which “appear” as a 25-35 year old woman to other valid end-users. At times, such software, scripts, or other machine-readable instruction sets are executed by the fraudulent end-user’s system. At other times, such software, scripts, or other machine-readable instruction sets are executed by a number of different systems (e.g., bots), each of which may operate autonomously or under the control of the fraudulent end-user to form a botnet.

[0008] The autonomous generation of fraudulent end-user profiles using a limited set of attribute values frequently results in the generation of a number of fraudulent end-user profiles having attribute values that fall into definable “clusters.” A fraud detection system having the ability to identify attribute values that define these clusters permits the matching service entity to both proactively prevent the registration of suspected fraudulent end-users having profiles that include attributes that have been found indicative of fraudulent end-user profiles as well as reactively remove from the matching service database registered suspected fraudulent end-users having profiles that include attributes that have

been found indicative of fraudulent end-user profiles. Such a system is also beneficially able to detect and respond to changes in fraudulent end-user behavior, for example a change in end-user attribute values including end-user personal attribute values as well as end-user system attribute values. In addition to fraudulent end-user profiles, other end-users profiles may also be deemed undesirable. Such undesirable user profiles may include user profiles logically associated with married end-users, user profiles logically associated with non-paying or delinquent paying end-users, and the like. A fraud detection system quantifies the likelihood that any particular attribute value or combination of attribute values appear in a fraudulent or otherwise undesirable end-user profile. Using this information, the fraud detection system determines a score or value indicative of the likelihood that a selected end-user profile is fraudulent. Based on the valuation of the respective end-user profile, the fraud detection system autonomously screens or removes a fraudulent end-user profile from addition to the matching service entity end-user database, autonomously adds or saves a valid end-user profile to the matching service entity end-user database, or forwards a suspected fraudulent end-user profile to a customer service representative for additional review.

[0009] Such fraud detection systems may obtain a number of known fraudulent end-user profiles, suspected fraudulent end-user profiles, and/or otherwise undesirable end-user profiles. Using statistical analysis techniques that include clustering the end-user profiles by attributes and attribute values and/or combinations of attributes and attribute values, the fraud detection system identifies on a continuous, periodic, or aperiodic basis those attribute values and/or attribute value combinations that appear in fraudulent end-user profiles. Using this data, the fraud detection system generates one or more queries to identify those end-user profiles having attribute values or combinations of attribute values that likely indicate a fraudulent end-user profile. The fraud detection system can run these queries against incoming registrations to identify and screen fraudulent end-user profiles from addition to the matching service entity’s end-user database and can also run these queries against such stored end-user profile databases to identify and remove fraudulent end-user profiles resident in the end-user database.

[0010] A method of operation in at least a portion of a system to detect at least one of accounts or related profiles suspected of being fraudulent, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, may be summarized as, for each of a plurality of profiles, computing by the at least one processor a representation of dissimilarity based at least in part on a respective attribute value of each of a plurality of attributes logically associated with the respective profile; performing a clustering of the profiles by the at least one processor, based on the representation of dissimilarity, the performance of the clustering resulting in a number of clusters, each cluster comprising one or more of the profiles; selecting clusters of the profiles above a threshold; in a first pass, for each of at least some of the selected clusters, identifying by the at least one processor each of a number of attributes and attribute value combinations that

occur frequently in the profiles of the respective cluster; in a second pass, for each of at least some of the selected clusters and for each attribute identified as occurring frequently in the first pass, identifying by the at least one processor one or more additional attribute values for the respective attribute that occur frequently in the profiles of the respective cluster; and for each of at least some of the selected clusters, preparing a respective query to identify the accounts or the related profiles suspected of being fraudulent based at least in part on the processor identified attribute and attribute value combinations and the processor identified one or more additional attribute values.

[0011] Computing a representation of dissimilarity may include, for each of the profiles, computing a dissimilarity matrix based on the respective values of each of the number of attributes logically associated with the profile. Computing a dissimilarity matrix may include computing the dissimilarity matrix via at least one of Manhattan distances, Euclidean distances or Gower coefficients. Computing a dissimilarity matrix may include computing the dissimilarity matrix based on at least one of: an identity of an Internet Service Provider (ISP) associated with the creation of the respective profile; an identity of an operating system used during a most recent login associated with the respective profile; one or more HTTP user string properties associated with the respective profile; a login associated with the respective profile; an identified gender associated with the respective profile; a cookie associated with the respective profile; a local machine time associated with the respective profile; an electronic mail (email) domain associated with the respective profile; a signup IP domain associated with the respective profile; one or more profile details associated with the respective profile; an IP country associated with the respective profile; or a time to create the respective profile. Performing a clustering of the profiles may include performing an agglomerative clustering of the profiles based on the representation of dissimilarity. Performing a clustering of the profiles may include forming combinations of profiles that minimize an average of dissimilarities between one cluster of profiles and another cluster of profiles. Performing a clustering of the profiles may include performing a clustering based on the dissimilarity matrices of the respective profiles. Selecting clusters of profiles above a threshold may include selecting clusters which include at least a threshold number of profiles. The performance of the clustering may result in a dendrogram with a top-most node representing all of the plurality of profiles and the bottom-most nodes representing respective individual ones of the profiles, a plurality of intermediate nodes between the top-most and the bottom-most nodes representing respective sets of profiles where the profiles in any set of profiles are similar to one another to a degree, and wherein selecting clusters of profiles above a threshold may include selecting clusters below a cut line in the dendrogram. Performing a clustering of the profiles may include performing an agglomerative nesting clustering of the profiles. Performing an agglomerative nesting clustering of the profiles may include performing an agglomerative nesting clustering of the profiles using at least one of: a nearest neighbor method, a furthest neighbor method, a pair group average method, a centroid method, a median method, and Ward's method. Performing a clustering of the profiles may include performing a divisive analysis clustering of the profiles. Identifying each of a number of attributes and attribute value combinations that

occur frequently in the profiles of the respective cluster may include identifying attributes and attribute value combinations that are above a minimum support threshold. Identifying any additional attribute values for the respective attribute that occur frequently in the profiles of the respective cluster may include identifying any additional attribute values that occur above a combine threshold.

[0012] The method of operation in at least a portion of a system to detect at least one of accounts or related profiles suspected of being fraudulent, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor may further include normalizing the attribute values for at least some of the attributes before performing the clustering of the profiles.

[0013] In the first pass identifying each of a number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster may include performing association rule learning on each of the at least some of the selected clusters. In the first pass identifying each of a number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster may include conducting a depth-first search for the combinations of attribute and attribute values that form a defined minimum percentage subset of the respective cluster. Conducting a depth-first search may include executing an association rule learning method on the profiles of the respective cluster that includes at least one of: an Éclat algorithm, an Apriori algorithm, an FP-growth algorithm, GUHA Procedure ASSOC, or an OPUS search. In the second pass, the method may include, for each identified additional value, adding the identified additional value to a set of identified values for the respective attribute. Preparing a respective query may include defining attribute values for each of a number of the attributes which are indicative of a pattern of fraudulently generated profiles, and may further include: uploading the queries to a production system, the production system comprising at least one processor that is separate and distinct from the at least one processor of the system.

[0014] The method of operation in at least a portion of a system to detect at least one of accounts or related profiles suspected of being fraudulent, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, may further include: identifying a subset of suspicious profiles from a set of profiles; and providing the subset of suspicious profiles to the at least one processor as the plurality of profiles.

[0015] Identifying the subset of suspicious profiles may include applying a semi-naïve Bayesian classification against the set of profiles.

[0016] The method of operation in at least a portion of a system to detect at least one of accounts or related profiles suspected of being fraudulent, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium commu-

nicatively coupled to the at least one processor, may further include: for each of at least some of the selected clusters, preparing a respective second query to identify the accounts or the related profiles suspected of being fraudulent based at least in part on the processor identified attribute and attribute value combinations and the processor identified one or more additional attribute values; and combining by the at least one processor the query and the second query to generate a third query to identify the accounts or the related profiles suspected of being fraudulent based at least in part on the processor identified attribute and attribute value combinations and the processor identified one or more additional attribute values.

[0017] A system to detect at least one of accounts or related profiles suspected of being fraudulent may be summarized as including: at least one processor; and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, wherein the at least one processor may: for each of a plurality of profiles, compute a representation of dissimilarity based at least in part on a respective attribute value of each of a plurality of attributes logically associated with the respective profile; perform a clustering of the profiles based on the representation of dissimilarity, which results in a number of clusters, each cluster comprising one or more of the profiles; select clusters of the profiles above a threshold; in a first pass, for each of at least some of the selected clusters, identify each of a number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster; in a second pass, for each of at least some of the selected clusters and for each attribute identified as occurring frequently in the first pass, identify one or more additional attribute values for the respective attribute that occur frequently in the profiles of the respective cluster; and for each of at least some of the selected clusters, prepare a respective query to identify the accounts or the related profiles suspected of being fraudulently generated based at least in part on the processor identified attribute and attribute value combinations and the one or more processor identified additional attribute values.

[0018] For each of the profiles, the at least one processor may compute a dissimilarity matrix based on the respective values of each of the number of attributes logically associated with the profile in order to compute the representation of dissimilarity. The at least one processor may compute the dissimilarity matrix via at least one of Manhattan distances, Euclidean distances or Gower coefficients. The at least one processor may compute the dissimilarity matrix based on at least one of: an identity of an Internet Service Provider (ISP) associated with the creation of the respective profile; an identity of an operating system used during a most recent login associated with the respective profile; a login associated with the respective profile; an identified gender associated with the respective profile; a cookie associated with the respective profile; an electronic mail (email) domain associated with the respective profile; a signup IP domain associated with the respective profile; one or more profile details associated with the respective profile; an IP country associated with the respective profile; or a time to create the respective profile. The at least one processor may form combinations of profiles that minimize a measure of dissimilarities between one cluster of profiles and another

cluster of profiles in order to perform the clustering of the profiles. The at least one processor may perform the clustering based on the dissimilarity matrices of the respective profiles. Clusters of profiles may be selected which include at least a threshold number of profiles. The performance of the clustering may result in a dendrogram with a top-most node representing all of the plurality of profiles and the bottom-most nodes representing respective individual ones of the profiles, a plurality of intermediate nodes between the top-most and the bottom-most nodes representing respective sets of profiles where the profiles in any set of profiles are similar to one another to a degree, and clusters below a cut line in the dendrogram are selected. The at least one processor may perform an agglomerative nesting clustering of the profiles based on the representation of dissimilarity. The at least one processor may perform an agglomerative nesting clustering of the profiles based on the representation of dissimilarity using at least one of: a nearest neighbor method, a furthest neighbor method, a pair group average method, a centroid method, a median method, and Ward's method. The at least one processor may perform a divisive analysis clustering of the profiles based on the representation of dissimilarity. The at least one processor may identify attributes and attribute value combinations that are above a minimum support threshold in order to identify each of the number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster. The at least one processor may identify any additional attribute values that occur above a combine threshold. The at least one processor may further normalize the attribute values for at least some of the attributes before the clustering of the profiles. The at least one processor may perform association rule learning on each of the at least some of the selected clusters, in order to identify each of the number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster in the first pass. The at least one processor may conduct a depth-first search for the combinations of attribute and attribute values that form a defined minimum percentage subset of the respective cluster, in order to identify each of the number of attributes and attribute value combinations that occur frequently in the profiles of the respective cluster in the first pass. The at least one processor may conduct a depth-first search that includes executing an association rule learning method on the profiles of the respective cluster that includes at least one of: an Éclat algorithm, an Apriori algorithm, an FP-growth algorithm, GUHA Procedure ASSOC, or an OPUS search. The at least one processor may add the identified additional value to a set of identified values for the respective attribute, as part of the second pass. The at least one processor may further upload the queries to a production system, the production system comprising at least one processor that is separate and distinct from the at least one processor of the system. The at least one processor may further receive a subset of suspicious profiles as the plurality of profiles, the subset of suspicious profiles selected from a set of profiles. The profiles may be profiles of individuals and the attributes values may include both self-reported attribute values and non-self-reported attribute values. The at least one processor may further, for each of at least some of the selected clusters, prepare a respective second query to identify the accounts or the related profiles suspected of being fraudulently generated based at least in part on the processor identified attribute and attribute value combinations and the processor identified one

or more additional attribute values; and combine the query and the second query to generate a third query to identify the accounts or the related profiles suspected of being fraudulently generated based at least in part on the processor identified attribute and attribute value combinations and the processor identified one or more additional attribute values.

[0019] A method of operation in at least a portion of a system to detect suspected fraudulently generated profiles, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, may be summarized as: computing, by the at least one processor, a value indicative of an initial probability of a profile in a set of profiles being at least one of a fraudulent profile $[p(S)]$ or a valid profile $[p(V)]$ based at least in part on historical profile data; computing, by the at least one processor, for an attribute or a combination of attributes associated with the profiles in the set of profiles, a respective value indicative of a likelihood of the attribute or the combination of attributes being associated with at least one of a fraudulent profile $[p(x_i|S)]$ or a valid profile $[p(x_i|V)]$; and for each profile in the set of profiles, computing, by the at least one processor, a value indicative of a conditional probability that the respective profile is one of either fraudulent or valid based on an attribute or a combination of attributes associated with the respective profile determined at least in part using the respective initial probabilities and likelihoods.

[0020] Computing a value indicative of an initial probability of profiles being at least one of a fraudulent profile $[p(S)]$ or a valid profile $[p(V)]$ may include computing both a value indicative of the initial probability of profiles being fraudulent $[p(S)]$ and a value indicative of the initial probability of profiles being valid $[p(V)]$.

[0021] Computing a value indicative of a conditional probability that the respective profile is fraudulent may include computing the value indicative of a conditional probability that the respective profile is fraudulent according to:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)}$$

[0022] Computing a value indicative of a likelihood of the attribute or the combination of attributes being associated with at least one of a fraudulent profile $[p(x_i|S)]$ or a valid profile $[p(x_i|V)]$ may include computing the respective value indicative of a respective likelihood based at least in part on: a Hyper Text Transfer Protocol (http) referrer associated with the respective profile; an Internet Service Provider (ISP) associated with the respective profile; an Internet Protocol (IP) of signup and last logins associated with the respective profile; a count of page requests made during a sign up for the respective profile; an identity of one or more plugins present in a browser associated with the respective profile; a location (state/province/location) associated with a sign up for the respective profile; an electronic mail (email) address associated with the respective profile; an http via associated with the respective profile; an identity of one or more fonts loaded on browser associated with the respective

profile; a most recent operating system (OS) associated with the respective profile; an http accept language associated with the respective profile; one or more cookies used to track individual computers associated with the respective profile; a username associated with the respective profile; a user age associated with the respective profile; a birth order associated with the respective profile; a number of siblings associated with the respective profile; one or more cookies that contain a user identifier of the most recent users to log in on using a given instance of a processor-based device; a self-identified country associated with the respective profile; an IP Country associated with the respective profile; one or more various screen properties associated with the respective profile; an identity of a browser used at signup associated with the respective profile; a self-identified profession associated with the respective profile; IP blocks of signup and at least two most recent logins associated with the respective profile; an IP country of at least two most recent visits associated with the respective profile; a time to create the respective profile; an electronic mail (email) domain associated with the respective profile; a self-identified postal or location code associated with the respective profile; one or more UTM cookies; or one or more user keystrokes and cut and paste properties used in creating the respective profile.

[0023] The method of operation in at least a portion of a system to detect suspected fraudulently generated profiles, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, may further include ranking the profiles into groups.

[0024] The method of operation in at least a portion of a system to detect suspected fraudulently generated profiles, the system which includes at least one processor and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, may further include providing result to a front end system for deletion, for clearance, or for investigation.

[0025] A fraudulent profile detection system to detect at least one of accounts or related profiles suspected of being fraudulently generated may be summarized as including: at least one processor; and at least one nontransitory processor-readable medium that stores at least one of processor-executable instructions or processor-executable data, the at least one nontransitory processor-readable medium communicatively coupled to the at least one processor, wherein the at least one processor may: compute a value indicative of an initial probability of a profile in a set of profiles being at least one of a fraudulent profile $[p(S)]$ or a valid profile $[p(V)]$ based at least in part on historical profile data; for an attribute or a combination of attributes associated with the profiles in the set of profiles, compute a respective value indicative of a likelihood of the attribute or the combination of attributes being associated with at least one of a fraudulent profile $[p(x_i|S)]$ or a valid profile $[p(x_i|V)]$; and for each profile in the set of profiles, compute a value indicative of a conditional probability that the respective profile is one of either fraudulent or valid based on an attribute or a combi-

nation of attributes associated with the respective profile determined at least in part using the respective initial probabilities and likelihoods.

[0026] The at least one processor may compute both a value indicative of the initial probability of profiles being fraudulent [p(S)] and a value indicative of the initial probability of profiles being valid [p(V)].

[0027] The at least one processor may compute a value indicative of a conditional probability that the respective profile is fraudulent includes computing the value indicative of a conditional probability that the respective profile is fraudulent according to:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)}$$

[0028] The at least one processor may further rank the profiles into groups using the respective value indicative of a conditional probability that the respective profile is fraudulent. The at least one processor may further generate at least one output logically associated with each profile, the at least one output indicative of at least one of the following: a deletion indicator, a clearance indicator, or a further investigation indicator.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0029] In the drawings, identical reference numbers identify similar elements or acts. The sizes and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not drawn to scale, and some of these elements are arbitrarily enlarged and positioned to improve drawing legibility. Further, the particular shapes of the elements as drawn are not intended to convey any information regarding the actual shape of the particular elements, and have been solely selected for ease of recognition in the drawings.

[0030] FIG. 1 is a schematic view of a networked matching service environment according to one illustrated embodiment, including a fraudulent profile detection system; and a plurality of end-user client entities of the matching service each with associated processor-based devices to provide communications with the matching service system.

[0031] FIG. 2 is a functional block diagram of a fraudulent profile detection system networked to one of the end-user client processor-based devices, according to one illustrated embodiment.

[0032] FIG. 3 is a dendrogram illustrating an example method useful for classifying end-user client entities based on one or more defined end-user attributes, according to one illustrated embodiment.

[0033] FIG. 4 is a flow diagram showing a high-level method of generating queries for use in a fraudulent profile detection system to detect client profiles meeting one or more fraudulent or otherwise undesirable profile characteristics, according to one illustrated embodiment.

[0034] FIG. 5 is a flow diagram showing a low-level method of generating a compound query that includes at least two individual queries for use by a fraud detections

system to detect suspected fraudulent or otherwise undesirable end-user profiles, according to one illustrated embodiment.

[0035] FIG. 6 is a flow diagram showing a low-level method of detecting fraudulent end-user profiles present in an end-user database of a matching service system, according to one illustrated embodiment.

[0036] FIG. 7 is a flow diagram showing a low-level method of detecting suspected fraudulent or otherwise undesirable end-user profiles based on an attributes and attribute values logically associated with an end-user profile provided to a matching service system, according to one illustrated embodiment.

[0037] FIG. 8 is a flow diagram showing a low-level method of determining a score or valuation indicative of the likelihood that an end-user profile is logically associated with a fraudulent or otherwise undesirable end-user based on the end-user attributes and attribute values associated with the profile, according to one illustrated embodiment.

[0038] FIG. 9 is a flow diagram showing a low-level method of generating one or more compound queries that include two or more simple queries developed using known or suspected fraudulent or otherwise undesirable end-user attributes and attribute values as machine learning training examples, according to one illustrated embodiment.

[0039] FIG. 10 is a flow diagram showing a low-level method of generating queries by using end-user profile data in a clustering analysis to detect attributes and attribute values appearing at frequencies greater than a defined threshold in known or suspected fraudulent or otherwise undesirable end-user profiles, according to one illustrated embodiment.

[0040] FIG. 11 is a flow diagram showing a low-level method of identifying attributes and attribute values characteristic of suspected fraudulent or otherwise undesirable end-user profiles by using end-user profile data in a clustering analysis to detect those attributes and attribute values appearing at frequencies greater than a defined threshold in known or suspected fraudulent or otherwise undesirable end-user profiles, according to one illustrated embodiment.

DETAILED DESCRIPTION

[0041] In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments.

[0042] However, one skilled in the relevant art will recognize that embodiments may be practiced without one or more of these specific details, or with other methods, components, materials, etc. In other instances, well-known structures associated with computer systems, server computers, and/or communications networks have not been shown or described in detail to avoid unnecessarily obscuring descriptions of the embodiments. In other instances, well-known mathematical and statistical methods for performing clustering analyses, dissimilarity analyses, and the like have not been described in detail to avoid unnecessarily obscuring descriptions of the embodiments.

[0043] Unless the context requires otherwise, throughout the specification and claims which follow, the word “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense, that is as “including, but not limited to.”

[0044] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular

feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0045] As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

[0046] The headings and Abstract of the Disclosure provided herein are for convenience only and do not interpret the scope or meaning of the embodiments.

[0047] FIG. 1 shows a networked environment 100 in which matching services are provided via a network 150 to a plurality of valid end-user client entities 120a-120n (only three shown, collectively “valid end-users 120”) and to a number of fraudulent or otherwise undesirable end-user client entities 130a-130n (only two shown, collectively “fraudulent end-users 130,” according to one illustrated embodiment. In at least some implementations, the matching services are provided by a matching service entity 102 that incorporates or is otherwise communicably coupled to one or more fraud detection system(s) 104.

[0048] Each of the valid end-users 120a-120n generates a respective profile 124a-124n (collectively, “valid end-user profile 124”) that is logically associated with the respective end-user 124 by the matching service entity 102. Each end-user profile 124a-124n includes data indicative of a number of attributes 126a-126n (collectively, “valid end-user attributes 126”), each having one or more attribute values associated therewith. In some instances, valid end-user attributes 126 may correspond to or be indicative of various end-user physical, mental, and/or emotional aspects. Some or all of such valid end-user attributes 126 may be acquired directly from the end-user 120 by the matching service entity (e.g., in the form of self-reported information, questionnaires, interviews, through examination of historical text messages, through examination of historical emails provided by the end-user client entity, and the like). An example of such a self-reported attribute and attribute value illustrated by the attribute “age” and the end-user selected associated attribute value of “45-55.” In some instances, some or all of such valid end-user attributes 126 may be acquired indirectly from the end-user (e.g., via personality profiles, based on examining historical interests of the end-user client, based on Websites visited by the end-user client, based on browsing habits of the end-user client, and the like).

[0049] Illustrative, non-exhaustive, examples of such self-reported end-user attributes include: an end-user generated username; an end-user supplied age; an end-user supplied profession; an end-user supplied email address/email domain; an end-user supplied state/province/location; an end-user supplied ZIP code/postal code/location code.

[0050] In some instances, the matching service entity 102 also acquires and stores any number of valid end-user attributes 126 that are autonomously provided by and/or associated with the end-user processor-based device 122 used to access the matching service entity 102. Some or all

of such information may be autonomously acquired from the end-user processor-based device 122 (IEEE 802 or “MAC” address, Network Interface Card “NIC” identification, operating system, system cookie data, plugins data, time required to create the end-user client, etc.). Some or all of such valid end-user attributes and attribute values 126 may be acquired from the internet service provider (“ISP”) and/or networks used by the end-user processor-based device 122 to access the matching service entity 102 (Internet Protocol “IP” address, ISP information, end-user client location information, email domain name, sign-up domain name, IP country, etc.).

[0051] Illustrative, non-exhaustive, examples of such autonomously collected end-user attributes include: a referring Website or URL location; the ISP; the IP address associated with an end-user signup and/or last login; the plugins present in the end-user’s browser; the http via; the end-user’s last operating system; cookies present on the end-user’s system; the country of the IP used by the end-user; the browser used by the end-user to signup; the IP blocks of the last two end-user sign-in events; the IP country of the last two end-user visits; the time required to create an end-user profile; and user keystrokes and cut and paste properties for creating an end-user profile.

[0052] Valid end-user profiles 124 along with the logically associated attribute(s) 126 and attribute value(s) are stored in a number of end-user data stores or databases 110 that are retained, stored, and/or maintained on nontransitory storage media 106a-106b (collectively, “nontransitory storage media 106”). The nontransitory storage media 106 may be communicably coupled to or integral with the matching service entity 102 and/or fraud detection system 104. In some instances, at least a portion of the nontransitory storage media 106a may be disposed local to the matching service entity 102 and/or fraud detection system 104. In some instances, at least a portion of the nontransitory storage media 106b may be disposed remote from the matching service entity 102 and/or fraud detection system 104 and communicably coupled via one or more networks 108 (e.g., local area networks “LANs;” wide area networks “WANs;” or worldwide networks such as the Internet).

[0053] Each of the fraudulent end-users 130a-130n generates a number of fraudulent end-user profiles 144a-144n (collectively, “fraudulent end-user profile 144”), each logically associated with a respective virtual end-user 140a-140n. Each fraudulent end-user profile 144a-144n includes a number of attributes 146a-146n (“collectively, fraudulent end-user attributes 146”) provided by the fraudulent end-user 130a-130n and/or the fraudulent end-user’s respective processor-based device 132a-132n. Fraudulent end-user profiles 144, along with the logically associated attributes 146 and attribute value(s) are also stored in the number of end-user data stores or databases 110 retained, stored, and/or maintained on the nontransitory storage media 106.

[0054] In some instances, fraudulent end-user 130a may execute machine readable instruction sets that cause a number of processor-based devices 148a-148n to each generate one or more virtual end-users 140, each having a fraudulent end-user profile 144 that contains a number of logically associated fraudulent end-user attributes 146 and attribute values. An example of this type of conduct is a “botnet” attack where the fraudulent end-user 130 installs software, malware, or similar machine-readable instruction sets on a number of usually unsuspecting processor-based devices

148a-148n which then function as bots in a botnet. Either autonomously or at the direction of the fraudulent end-user **130a**, each of these processor-based devices **148** will attempt to generate and register with the matching service entity **102** any number of virtual end-users **140a-140n**, each logically associated with a respective fraudulent end-user profile **144a-144n**.

[0055] In other instances, a processor-based device **132n** operated by a fraudulent end-user **130n** may execute one or more machine readable instruction sets that cause the processor-based device **132n** to generate a number of virtual end-users **140a-140n**, each having a respective, logically associated, fraudulent end-user profile **144a-144n** having associated therewith a number of fraudulent end-user attributes **146a-146n** and attribute values. An example of this type of conduct is where the fraudulent end-user **130** installs software or similar machine-readable instruction sets on the processor-based device **132n**. The processor-based device **132n** then functions as a bot, generating any number of virtual end-users **140a-140n**, each logically associated with a respective fraudulent end-user profile **144a-144n**. In at least some instances, such activity is performed by the fraudulent end-user through a series of proxies or proxy servers to mask the identity of the fraudulent end-user **130** and/or the fraudulent end-user's processor-based device **132**, for example by masking processor-based device attribute values such as Internet Service Provider ("ISP") information, Internet Protocol ("IP") address, and/or physical location of the fraudulent end-user **130**.

[0056] In some instances, the matching service entity **102** inadvertently or unknowingly acquires and stores any number of fraudulent end-user attributes **146** and attribute values that are autonomously provided by and/or associated with the fraudulent end-user processor-based device **132** (or by one or more proxy processor-based devices **148** infected by malware supplied by the fraudulent end-user **130**). Some or all of these attributes and attribute values may be autonomously acquired from the fraudulent end-user processor-based device **132** (MAC address, NIC identification, operating system, system cookie data, plugins data, time required to create the end-user client, etc.). Some or all of such fraudulent end-user attributes **146** may be acquired from the Internet service provider ("ISP") and/or networks used by the fraudulent end-user processor-based device **132** to access the matching service entity **102** (Internet Protocol "IP" address, ISP information, end-user client location information, email domain name, sign-up domain name, IP country, etc.).

[0057] Fraudulent end-users **130** may attempt to initiate contact with valid end-users **120** via the matching services provided by the matching service entity **102**. Additionally or alternatively, various processor-based devices operating as bots may attempt to initiate the contact. Such social networking services and/or matching services may include but are not limited to matching services for the supply or exchange of goods and/or services; matching services for personal friendships; matching services for individuals sharing common goals, aspirations, and/or interests; and matching services for individuals seeking romantic relationships. Once a relationship is established between a valid end-user **120** and the virtual end-user **140**, the fraudulent end-user **130** is able to communicate with the valid end-user **120** through the virtual end-user **140**. Such communication may occur through the matching service entity **102** or external to

the matching service entity **102** (e.g., via email, text message, letter, and/or telephone). In some instances, the fraudulent end-user **130** can then extract personal information and/or financial information from the valid end-user **120** thereby exposing the valid end-user to activities such as fraud (e.g., an overseas love interest that requires money to travel or for a sick relative) and identity theft (e.g., the valid end-user's financial information is needed to acquire travel documents).

[0058] In some instances, the fraudulent end-user **130** may use the fraudulent end-user processor-based device **132n** and/or one or more malware infected processor-based devices **148** to collect information from valid end-users **120** of the matching service entity **102**. For example, either autonomously or at the direction of a fraudulent end-user **130**, the fraudulent end-user processor-based device **132n** or one or more proxy processor-based devices **148** may autonomously message, communicate, or otherwise electronically exchange information with one or more valid end-users **120**. Upon receiving a response to the autonomously generated message or communication from the valid end-user, the fraudulent end-user processor-based device **132n** or one or more proxy processor-based devices **148** may scrape or otherwise extract information (e.g., email information, telephone number information, address information) from the response.

[0059] In some instances, the fraudulent end-user **130** may use the fraudulent end-user processor-based device **132n** and/or one or more malware infected processor-based devices **148** to cause information or messages to be directed to a valid end-user **120**. For example, either autonomously or at the direction of a fraudulent end-user **130**, the fraudulent end-user processor-based device **132n** or one or more proxy processor-based devices **148** may autonomously generate one or more messages to one or more valid end-users **120**. When a valid end-user responds to the message, the fraudulent end-user processor-based device **132n** and/or one or more malware infected processor-based devices **148** ignore the response, thereby leading the valid end-user **120** to believe that a rejection of their expressed interest has been rejected. Over time, such rejections can create an online environment where valid end-users **120** become frustrated with the perceived lack of response from what the valid end-user believes to be other "end-users."

[0060] Fraudulent end-users **130** may use statistical information or other insight to target a particular group or segment of the population of valid end-users **120**. For example, a fraudulent end-user **130** may determine that men over 50 years of age are most susceptible to engaging in relationships with women between the ages of 25 and 35 via the matching service entity **120**. Using this information, the fraudulent end-user **130** may create or generate malware or other machine-readable instruction sets that cause one or more proxy processor-based devices **148** to either autonomously or at the direction of the fraudulent end-user **130**, generate a number of virtual end-users **140a-140n**, each having a respective, logically associated, fraudulent end-user profile **144a-144n** containing attributes **146a-146n** and attribute values corresponding to a 25-35 year old woman. Thus, the fraudulent end-user profiles **144a-144n** generated by a fraudulent end-user **130** will each have logically associated a group fraudulent end-user attributes **146**. The attribute values associated with each of the fraudulent end-user attributes may be selected from one or more defined

groups of attribute values. For example, if the fraudulent end-user profile **144** includes the attribute “hair color,” the fraudulent end-user **130** may define a set of attribute values consisting of “blonde, brunette, and black” from which one attribute value is selected at random when generating a fraudulent end-user profile **146**. Although an end-user profile may include a number of attributes, the fact that the attribute values are selected from a defined and/or targeted set of attribute values selected by the fraudulent end-user **130** effectively limits the available number of attribute value permutations. Thus, each of the fraudulent end-user profiles **144a-144n** included within the population of fraudulent profiles generated by a fraudulent end-user **130** will demonstrate at least a limited similarity of attributes **146** or parameter combinations (female, 25-35 years old, IP address of profile generation, date and time of profile generation, etc.). In contrast, the diversity existent in the population of valid end-users **120** generally results in dissimilar or a broader and/or more varied distribution of valid end-user attributes **126** across a population of valid end-user profiles **124**.

[0061] The success of a matching service ultimately depends upon the perceived value of the service to valid end-users **120**. Successful pairings resulting in favorable exchanges of goods and/or services, friendships, groups of individuals sharing common interests, and long-term romantic relationships add to the perceived value of the matching service to valid end-users. Conversely, fraud, identity theft, and the overall feeling of loss and mistrust engendered by the activities of fraudulent end-users **130** rapidly detracts from the perceived value of the matching service to valid end-users **120**. Thus, the detection and prevention of fraudulent end-user sign-ups and the prompt and ongoing detection and removal of fraudulent end-user profiles from end-user data stores or databases provides an important step in improving the perceived and actual value of the matching service entity **102**.

[0062] The fraud detection system **104** provides the matching service entity **102** with the ability to autonomously, semi-autonomously, or manually “learn” the end-user attributes **146** and attribute values and/or combinations of end-user attributes and attribute values that are most frequently found in fraudulent end-user profiles **144**. In at least some implementations, the fraud detection system **104** may also provide the matching service entity **102** with the ability to autonomously, semi-autonomously, or manually “learn” the end-user attributes **126** and attribute values and/or combinations of end-user attributes and attribute values most frequently found in valid end-user profiles **124**. Advantageously, in addition to identifying these attribute and attribute value patterns, the fraud detection system **104** provides a mechanism for reliability and at least semi-autonomously distinguishing between valid end-user profiles **124** and fraudulent end-user profiles **144**.

[0063] One or more programs or similar machine-readable instruction sets **112** that provide either or both machine learning and/or fraud detection capabilities are stored, retained, or otherwise maintained on the nontransitory storage media **106**. In some instances, one or more programs or machine-readable instruction sets may provide statistical analysis capabilities useful for detecting patterns, distributions, and/or frequencies of attributes **146** and attribute values and/or combinations of attributes **146** and attribute values in both historical fraudulent end-user profiles **144** as

well as current suspected fraudulent end-user profiles **144**. In some instances, one or more programs or machine-readable instruction sets may provide machine learning capabilities for the fraud detection system **104** using both historical fraudulent end-user profiles **144** as well as current suspected fraudulent end-user profiles **144** as training examples. Through statistical analysis and machine learning via known or suspected fraudulent end-user profile **144** training examples, the fraud detection system **104** can screen incoming end-user profiles prior to addition to the end-user data store or database **110** and/or screen the end-user data store or database **110** to detect, remove, or mark for removal existing fraudulent and suspected fraudulent end-user profiles **144**.

[0064] In some instances, the fraud detection system **104** generates a “score” or other similar valuation for each end-user profile based on the end-user attributes and attribute values and/or combinations of end-user attributes and attribute values associated with the respective end-user profile. In some implementations, the fraud detection system **104** uses the score or other valuation associated with each end-user profile to rank the profile into one of any number of groups. Such groups may include, but are not limited to, groups of end-user profiles classified as: fraudulent, suspected as being fraudulent, suspected as being valid, valid, or requiring additional investigation prior to determination of fraud. In some implementations, such profile scores or valuations may be provided to one or more customer service representatives or organizations for manual follow-up prior to the determination that the respective end-user profile is either a valid end-user profile **124** or a fraudulent end-user profile **144**. In some implementations, the fraud detection system **104** may autonomously prevent the storage of known fraudulent end-user profiles **144** in the end-user data store or database **110** and/or may remove known fraudulent end-user profiles **144** from the end-user data store or database.

[0065] The fraud detection system **104** includes a query generation sub-system that generates a number of queries based in whole or in part on known or suspected fraudulent end-user attributes **146** and attribute values and/or combinations of known or suspected fraudulent end-user attributes **146** and attribute values. In some instances, the query generation sub-system may be implemented in whole or in part in the form of one or more machine-readable instruction sets executed by the fraud detection system **104**.

[0066] The fraud detection system **104** additionally includes machine-readable instruction sets **112** that cause the fraud detection system to screen via one or more simple queries and/or one or more compound queries some or all of the incoming end-user profiles for end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that are associated with known or suspected fraudulent end-user profiles **144**. Responsive to detecting an incoming known or suspected fraudulent end-user profile **144**, the fraud detection system may autonomously reject the known or suspected fraudulent end-user **130** logically associated with the known or suspected fraudulent end-user profile **144**.

[0067] The machine-readable instruction sets **112** may additionally include one or more screening algorithms that cause the fraud detection system **104** to scan via one or more simple queries and/or one or more compound queries all or a portion of the end-user profiles stored, retained, or otherwise maintained in the end-user data store or database **110**

for end-user attributes **146** and attribute values and/or combinations of end-user attributes **146** and attribute values that are associated with known or suspected fraudulent end-user profiles **144**. Responsive to detecting a stored known or suspected fraudulent end-user profile **144**, the fraud detection system may autonomously quarantine and/or remove the known or suspected fraudulent end-user profile **144** from the end-user data store or database **110**.

[0068] While FIG. 1 illustrates a representative networked environment **100**, typical networked environments may include many additional computer systems and entities. The concepts taught herein may be employed in a similar fashion with more populated networked environments than that illustrated. FIG. 2 and the following discussion provide a brief, general description of a suitable networked matching service environment **200** in which the various illustrated fraud detection systems **104** might be implemented. Although not required, the embodiments will be described in the general context of computer-executable instructions, such as program application modules, objects, or macros stored on computer-or processor-readable media and executed by a computer or processor. Those skilled in the relevant art will appreciate that the illustrated embodiments, as well as other embodiments, can be practiced with other system configurations and/or other computing system configurations, including hand-held devices, multiprocessor systems, microprocessor-based or programmable consumer electronics, personal computers (“PCs”), networked PCs, mini computers, mainframe computers, and the like. The embodiments can be practiced in distributed computing environments where tasks or modules are performed by remote processing devices, which are linked through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices or media.

[0069] FIG. 2 shows a networked matching service environment **200** comprising one or more fraud detection systems **104** (only one illustrated) and one or more associated nontransitory machine-readable storage media **106** (only one illustrated). The associated nontransitory computer-or processor-readable storage media **106** is communicatively coupled to the fraud detection system(s) **104** via one or more communications channels, for example one or more parallel cables, serial cables, or wireless network channels capable of high speed communications, for instance via Universal Serial Bus (“USB”) 3.0 or via Thunderbolt®.

[0070] The networked matching service environment **200** also includes one or more valid end-user processor-based devices **122** (only one illustrated) and one or more fraudulent end-user processor-based devices **132** (only one illustrated). The valid end-user processor-based devices **122** and the fraudulent end-user processor-based devices **132** are communicably coupled to the fraud detection system(s) **104** by one or more communications channels, for example one or more wide area networks (WANs) **150**, for instance the Internet using Internet protocols. In operation, the valid end-user processor-based device **122** and the fraudulent end-user processor-based device **132** typically represent one or more valid end-user processor-based devices **122** or one or more fraudulent end-user processor-based devices **132** (e.g., desktop computer, laptop computer, ultraportable computer, tablet computer, smartphone, wearable computer) that are directly communicably coupled or indirectly communicably coupled through the matching system entity **102** to the

fraud detection system **104** via the one or more networks **150**. Although the fraud detection system **104** is described herein in the context of a standalone system, the fraud detection system **104** may, in fact, constitute only a portion of the matching service entity **102** (e.g., a sub-system installed in or communicably coupled to the matching service entity **102**). In such instances, components, sub-systems, and resources described forming a portion of the fraud detection system **104** should be understood as components, sub-systems, and resources that are shared with the matching service entity **102** or one or more common systems and/or resources that are allocated between the fraud detection system(s) **104** by the matching service entity **102**.

[0071] The networked matching service environment **200** may employ other computer systems and network equipment, for example additional servers, proxy servers, firewalls, routers and/or bridges. The fraud detection system(s) **104** will at times be referred to in the singular herein, but this is not intended to limit the embodiments to a single device since in typical embodiments there may be more than one fraud detection system(s) **104** involved. Unless described otherwise, the construction and operation of the various blocks shown in FIG. 2 are of conventional design. As a result, such blocks need not be described in further detail herein, as they will be understood by those skilled in the relevant art.

[0072] The fraud detection system(s) **104** may include one or more processing units **212a**, **212b** (collectively **212**), a system memory **214** and a system bus **216** that couples various system components including the system memory **214** to the processing units **212**. The processing units **212** may be any logic processing unit, such as one or more central processing units (CPUs) **212a**, digital signal processors (DSPs) **212b**, application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), etc. The system bus **216** can employ any known bus structures or architectures, including a memory bus with memory controller, a peripheral bus, and/or a local bus. The system memory **214** includes read-only memory (“ROM”) **218** and random access memory (“RAM”) **220**. A basic input/output system (“BIOS”) **222**, which can form part of the ROM **218**, contains basic routines that help transfer information between elements within the fraud detection system **104**, such as during start-up.

[0073] The fraud detection system **104** may include a hard disk drive **224** for reading from and writing to a hard disk **226**, an optical disk drive **228** for reading from and writing to removable optical disks **232**, and/or a magnetic disk drive **230** for reading from and writing to magnetic disks **234**. The optical disk **232** can be a CD-ROM, while the magnetic disk **234** can be a magnetic floppy disk or diskette. The hard disk drive **224**, optical disk drive **228** and magnetic disk drive **230** may communicate with the processing unit **212** via the system bus **216**. The hard disk drive **224**, optical disk drive **228** and magnetic disk drive **230** may include interfaces or controllers (not shown) coupled between such drives and the system bus **216**, as is known by those skilled in the relevant art. The drives **224**, **228** and **230**, and their associated computer-readable media **226**, **232**, **234**, provide nonvolatile storage of computer-readable instructions, data structures, program modules and other data for the master node server computer system **202**. Although the depicted fraud detection system(s) **104** is illustrated employing a hard disk **224**, optical disk **228** and magnetic disk **230**, those skilled in the

relevant art will appreciate that other types of computer-readable media that can store data accessible by a computer may be employed, such as WORM drives, RAID drives, magnetic cassettes, flash memory cards, digital video disks (“DVD”), Bernoulli cartridges, RAMs, ROMs, smart cards, etc.

[0074] Program modules can be stored in the system memory 214, such as an operating system 236, one or more application programs 238, other programs or modules 240 and program data 242. Application programs 238 may include instructions that cause the processor(s) 212 to identify attributes 146 and attribute values and/or combinations of attributes 146 and attribute values indicative of known and/or suspected fraudulent end-user profiles 144 using one or more classification systems. In at least some implementations, the fraud detection system 104 may be trained using known historical fraudulent end-user profiles 144 or historical suspected fraudulent end-user profiles 144 to provide a naïve or semi-naïve Bayesian classification system useful for identifying suspected fraudulent end-user profiles 144. In at least some implementations, the end-user attributes 146 included in the analysis of the known historical fraudulent end-user profiles 144 or historical suspected fraudulent end-user profiles 144 includes, but is not limited to: the ISP, the last login operating systems, the password properties of the end-user, end-user processor-based system plugins, end-user gender, end-user email domain information, end-user signup IP domain information, end-user profile details, IP country, the time required to generate the end-user profile, and similar. Application programs 238 may additionally include one or more machine-readable instruction sets that cause the processor(s) 212 to determine data indicative of one or more aspects of the dissimilarity existent in a set of end-user profiles. The set of end-user profiles may include any number of known fraudulent end-user profiles 146, any number of known valid end-user profiles 126, or any number of known valid end-user profiles 124 and known fraudulent end-user profiles 144. In at least some instances, the processor(s) 212 generate data indicative of a representation of the dissimilarity in a set of end-user profiles using a dissimilarity matrix. In at least some instances, the processor(s) 212 determine the dissimilarity matrix using at least one of: Manhattan distances, Euclidean distances, or Gower coefficients.

[0075] Application programs 238 may additionally include one or more machine-readable instruction sets that cause the processor(s) 212 to apply one or more clustering algorithms to a set of valid and/or fraudulent end-user profiles to provide a number of clusters of related valid and/or fraudulent end-user profiles. Each of the number of clusters includes at least one valid or fraudulent end-user profile.

[0076] In at least some implementations, the clustering algorithm may include one or more agglomerative clustering algorithms. In at least some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on a determined representation of dissimilarity in the set of valid and/or fraudulent end-user profiles. In at least some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on minimizing an average of dissimilarities between one determined user-profile cluster and another determined user profile cluster. In at least some implementations, the clustering algorithm may

include an agglomerative clustering algorithm that is based at least in part on clustering valid and/or fraudulent end-user profiles based at least in part on the dissimilarity matrices of the respective end-user profiles. In some instances, the clustering algorithm performed by the processor(s) 212 may include an agglomerative nesting clustering algorithm that clusters valid and/or fraudulent end-user profiles using at least one of: a nearest neighbor method, a furthest neighbor method, a pair group average method, a centroid method, a median method, and Ward’s method. In some instances, the clustering algorithm performed by the processor(s) 212 may include a divisive analysis clustering of valid and/or fraudulent end-user profiles. In some instances, the clustering algorithm performed by the processor(s) 212 may include normalizing the values associated with one or more valid and/or fraudulent end-user attributes and attribute values and/or combinations of valid and/or fraudulent end-user attribute and/or attribute values prior to performing the cluster analysis.

[0077] In at least some implementations, the clustering performed by the fraud detection system 104 can provide clustering data in the form of a tree or dendrogram such as the illustrative dendrogram 300 depicted in FIG. 3. In the dendrogram 300, individual end-users 302 are depicted at the bottom and the top 304 includes the entire set of end-users included in the cluster analysis performed by the fraud detection system 104. Between the top and bottom of the dendrogram, end-users are grouped in intermediate level clusters 306a-306e based on the valid and/or fraudulent end-user attributes logically associated with the respective valid and/or fraudulent end-users. Thus, the end-users included in each intermediate level cluster 306 will share a number of similarities. The number of similarities shared by the end-users within a particular cluster tends to decrease towards the top of the dendrogram 300 and tends to increase towards the bottom of the dendrogram 300.

[0078] Returning to FIG. 2, application programs 238 may additionally include one or more machine-readable instruction sets that cause the processor(s) 212 to autonomously or semi-autonomously (i.e., at least partially manually) select various clusters resulting from the clustering algorithm as including a number of suspected fraudulent end-user profiles 144. For example, using the dendrogram 300 depicted in FIG. 3, in at least some implementations, the processor(s) 212 can select a horizontal “cut” across the dendrogram 300. Intermediate level clusters 306 falling below the horizontal cut are discarded. Thus, a cut at line 308 provides four clusters, a cut at line 310 provides six clusters, and a cut at line 312 provides twenty-two clusters. Cutting the dendrogram 300 higher (e.g., at 308) results in a fewer number of clusters that include larger numbers of end-users 120, 140 that share relatively smaller number of similarities while cutting the dendrogram 300 lower (e.g., at 312) results in a greater number of clusters that include fewer numbers of valid and/or fraudulent end-users that share a relatively greater number of similarities. Cluster generation and selection algorithms other than the dendrogram discussed in detail above may be substituted.

[0079] Application programs 238 may additionally include one or more machine-readable instruction sets that cause the processor(s) 212, in a first pass, to identify respective end-user attribute values and/or combinations of end-user attribute values appearing above a first defined threshold in each of the clusters 306 suspected of including

a number of suspected fraudulent end-user profiles **144**. In at least some implementations, end-user attribute values and/or combinations of end-user attribute values occurring within a cluster **306** at a frequency above a first defined threshold (e.g., greater than “x” percent of the population of the respective cluster) are flagged or otherwise identified by the processor(s) **212** as indicative of the existence of a fraudulent end-user profile **144**. In at least some implementations, the processor(s) **212** use one or more algorithms to perform, at least in part, the depth-first search for frequent end-user attribute values and/or combinations of end-user attribute values that occur with a frequency exceeding the first defined threshold (i.e., the “support threshold”) in the subset of valid and/or fraudulent end-user profiles included in each of the clusters **306**. The algorithm used by the processor(s) **212** to perform the depth-first search for frequent end-user attribute values and/or combinations of end-user attribute values that occur with a frequency exceeding the first defined threshold can include at least one of: an Eclat algorithm, an Apriori algorithm, an FP-growth algorithm, a GUHA Procedure ASSOC, or an OPUS search.

[0080] Application programs **238** may additionally include one or more machine-readable instruction sets that cause the processor(s) **212**, in a second pass, to identify additional end-user attribute values and/or combinations of end-user attribute values appearing in the end-user attributes identified in the first pass and occurring with a frequency above a second defined threshold in each of the clusters **306**. In at least some implementations, end-user attribute values and/or combinations of end-user attribute values occurring within a cluster **306** at a frequency above a second defined threshold (e.g., greater than “y” percent of the population of the respective cluster) are flagged or otherwise identified by the processor(s) **212** as indicative of the existence of a fraudulent end-user profile **144**. In at least some implementations, the processor(s) use an algorithm to perform, at least in part, the search for frequent end-user attribute values and/or combinations of end-user attribute values that occur with a frequency exceeding the second defined threshold (i.e., the “combine threshold”) in the subset of fraudulent and/or suspected fraudulent end-user profiles included in each of the clusters **306**.

[0081] Application programs **238** may additionally include one or more machine-readable instruction sets that cause the processor(s) **212**, in a second pass, to generate one or more queries to identify fraudulent or suspected fraudulent end-user profiles **144** based on the presence of end-user attribute values **146** and/or combinations of end-user attribute values **146** identified in the first and second pass analyses appearing in the respective end-user profile **144**. Application programs **238** may additionally include one or more machine-readable instruction sets that cause the processor(s) **212** to combine one or more queries into a single compound query to identify fraudulent end-user profiles **144** based at least in part on the presence of including fraudulent end-user attributes **146** and attribute values and/or combinations of fraudulent end-user attributes **146** and attribute values.

[0082] In at least some implementations, the processor(s) **212** may run one or more queries against incoming or newly received end-user profiles to identify those end-user profiles including attribute values indicative of a fraudulent end-user profile or a suspected fraudulent end-user profile prior to adding the end-user profile to the end-user data store or

database **110**. In at least some implementations, the processor(s) **212** may run one or more queries against the valid and/or fraudulent end-user profiles included in the end-user data store or database **110** to identify fraudulent end-user profiles **144** based at least in part on the presence of including fraudulent end-user attributes **146** and attribute values and/or combinations of fraudulent end-user attributes **146** and attribute values.

[0083] While shown in FIG. 2 as being stored in the system memory **214**, the operating system **236**, application programs **238**, other programs/modules **240**, program data **242** and browser **244** can be stored on the hard disk **226** of the hard disk drive **224**, the optical disk **232** of the optical disk drive **228** and/or the magnetic disk **234** of the magnetic disk drive **230**.

[0084] An operator can enter commands and information into the fraud detection system **104** through input devices such as a touch screen or keyboard **246** and/or a pointing device such as a mouse **248**, and/or via a graphical user interface. Other input devices can include a microphone, joystick, game pad, tablet, scanner, etc. These and other input devices are connected to one or more of the processing units **212** through an interface **250** such as a serial port interface that couples to the system bus **216**, although other interfaces such as a parallel port, a game port or a wireless interface or a universal serial bus (“USB”) can be used. A monitor **252** or other display device is coupled to the system bus **216** via a video interface **254**, such as a video adapter. The fraud detection system(s) **104** can include other output devices, such as speakers, printers, etc.

[0085] The fraud detection system(s) **104** can operate in a networked environment using logical connections to one or more remote computers and/or devices. For example, the fraud detection system(s) **104** can operate in a networked environment using logical connections to one or more valid end-user processor-based devices **122** and to one or more fraudulent end-user processor-based devices **132**. Communications may be via tethered, wired, and/or wireless network architecture, for instance wired and wireless enterprise-wide computer networks, intranets, extranets, and/or the Internet. Other embodiments may include other types of communications networks including telecommunications networks, cellular networks, paging networks, and other mobile networks. There may be any variety of computers, switching devices, routers, bridges, firewalls and other devices in the communications paths between the fraud detection system(s) **104**, the one or more valid end-user processor-based devices **122** and the one or more fraudulent end-user processor-based devices **132**.

[0086] The one or more valid end-user processor-based devices **122** and the one or more fraudulent end-user processor-based devices **132** typically take the form of end-user processor-based devices, for instance personal computers (e.g., desktop or laptop computers), netbook computers, tablet computers, smartphones, portable computers, wearable computers, and other processor-based devices capable of executing machine-readable instruction sets. These end-user processor-based devices may be communicatively coupled to one or more routers and/or one or more server computers. For instance, the end-user processor-based devices **122**, **132** may be communicatively coupled through one or more firewalls to an Internet Service Provider or ISP.

[0087] The valid end-user processor-based devices **122** and the fraudulent end-user processor-based devices **132**

may include one or more processing units **268a**, **268b** (collectively **268**), system memories **269a**, **269b** (collectively **269**) and a system bus (not shown) that couples various system components including the system memory **269** to the processing unit **268**. The valid end-user processor-based devices **122** and the fraudulent end-user processor-based devices **132** will at times each be referred to in the singular herein, but this is not intended to limit the embodiments to a single valid end-user processor-based device **122** and/or a single fraudulent end-user processor-based device **132**. In typical embodiments, there will likely be a large number of valid end-user processor-based devices **122** and there may be more than one fraudulent end-user processor-based devices **132**.

[0088] The processing unit **268** may be any logic processing unit, such as one or more central processing units (CPUs), digital signal processors (DSPs), application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), etc. Non-limiting examples of commercially available computer systems include, but are not limited to, the Celeron, Core, Core 2, Itanium, and Xeon families of microprocessors offered by Intel® Corporation, U.S.A.; the K8, K10, Bulldozer, and Bobcat series microprocessors offered by Advanced Micro Devices, U.S.A.; the A5, A6, and A7 series microprocessors offered by Apple Computer, U.S.A.; the Snapdragon series microprocessors offered by Qualcomm, Inc., U.S.A.; and the SPARC series microprocessors offered by Oracle Corp., U.S.A.. Unless described otherwise, the construction and operation of the various blocks of the end-user processor based devices **122**, **132** shown in FIG. 2 are of conventional design. As a result, such blocks need not be described in further detail herein, as they will be understood by those skilled in the relevant art.

[0089] The system bus can employ any known bus structures or architectures, including a memory bus with memory controller, a peripheral bus, and a local bus. The system memory **269** includes read-only memory (“ROM”) **270a**, **270b** (collectively **270**) and random access memory (“RAM”) **272a**, **272b** (collectively **272**). A basic input/output system (“BIOS”) **271a**, **271b** (collectively **271**), which can form part of the ROM **270**, contains basic routines that help transfer information between elements within the end-user computer systems **206**, **208**, such as during start-up.

[0090] The valid end-user processor-based devices **122** and the fraudulent end-user processor-based devices **132** may also include one or more media drives **273a**, **273b** (collectively **273**), e.g., a hard disk drive, magnetic disk drive, WORM drive, and/or optical disk drive, for reading from and writing to computer-readable storage media **274a**, **274b** (collectively **274**), e.g., hard disk, optical disks, and/or magnetic disks. The computer-readable storage media **274** may, for example, take the form of removable media. For example, hard disks may take the form of a Winchester drives, and optical disks can take the form of CD-ROMs, while magnetic disks can take the form of magnetic floppy disks or diskettes. The media drive(s) **273** communicate with the processing unit **268** via one or more system buses. The media drives **273** may include interfaces or controllers (not shown) coupled between such drives and the system bus, as is known by those skilled in the relevant art. The media drives **273**, and their associated computer-readable storage media **274**, provide nonvolatile storage of computer readable instructions, data structures, program modules and

other data for the fraudulent end-user processor-based devices **132** and/or the valid end-user processor-based devices **122**. Although described as employing computer-readable storage media **274** such as hard disks, optical disks and magnetic disks, those skilled in the relevant art will appreciate that end-user computer systems **206**, **208** may employ other types of computer-readable storage media that can store data accessible by a computer, such as magnetic cassettes, flash memory cards, digital video disks (“DVD”), Bernoulli cartridges, RAMs, ROMs, smart cards, etc. Data or information, for example, electronic or digital documents or files or data (e.g., metadata, ownership, authorizations) related to such can be stored in the computer-readable storage media **274**.

[0091] Program modules, such as an operating system, one or more application programs, other programs or modules and program data, can be stored in the system memory **269**. Program modules may include instructions for accessing a Website, extranet site or other site or services (e.g., Web services) and associated WebPages, other pages, screens or services hosted by the insurance management system **102**. Program modules may include instructions for storing certain or selected electronic correspondence and/or electronic or digital documents or files or changes thereto to nontransitory computer-or processor readable storage medium, such as local media **274a**, **274b**. Alternatively, the instructions may cause retrieval of electronic correspondence and/or electronic or digital documents or files or changes to existing electronic correspondence and/or electronic or digital documents or files. Program modules may additionally include instructions for handling security such as ownership, password or other access protection and communications encryption.

[0092] The system memory **269** may also include other communications programs, for example a Web client or browser that permits the valid end-user processor-based devices **122** and the fraudulent end-user processor-based devices **132** to access and exchange data with sources such as Web sites of the Internet, corporate intranets, extranets, or other networks. The browser may, for example be markup language based, such as Hypertext Markup Language (HTML), Extensible Markup Language (XML) or Wireless Markup Language (WML), and may operate with markup languages that use syntactically delimited characters added to the data of a document to represent the structure of the document.

[0093] While described as being stored in the system memory **269**, the operating system, application programs, other programs/modules, program data and/or browser can be stored on the computer-readable storage media **274** of the media drive(s) **273**. An operator can enter commands and information into the fraudulent end-user processor-based devices **132** and the valid end-user processor-based devices **122** via a user interface **275a**, **275b** (collectively **275**) through input devices such as a touch screen or keyboard **276a**, **276b** (collectively **276**) and/or a pointing device **277a**, **277b** (collectively **277**) such as a mouse. Other input devices can include a microphone, joystick, game pad, tablet, scanner, etc. These and other input devices are connected to the processing unit **269** through an interface such as a serial port interface that couples to the system bus, although other interfaces such as a parallel port, a game port or a wireless interface or a universal serial bus (“USB”) can be used. A display or monitor **278a**, **278b** (collectively **278**) may be

coupled to the system bus via a video interface, such as a video adapter. The satellite node server computer system 206 can include other output devices, such as speakers, printers, etc.

[0094] FIG. 4 shows a high-level method 400 of generating queries for identifying fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 by a fraud detection system 104 such as that described in detail with regard to FIGS. 1-3, according to one illustrated embodiment. In at least some implementations, the fraud detection system 104 can autonomously, semi-autonomously, or manually identify any number of known fraudulent end-user profiles 144 and any number of suspected fraudulent end-user profiles 144. Each of the identified fraudulent and suspected fraudulent end-user profiles 144 include any number of logically associated attributes 146, each having at least one attribute value associated therewith. Using the attributes 146 and attribute values logically associated with the identified fraudulent and suspected fraudulent end-user profiles 144 as training examples, the fraud detection system 104 autonomously generates or semi-autonomously generates (e.g., autonomously generates queries that are approved by a customer service representative) a number of queries, each of which can be run against either an incoming or stored end-user profile to determine whether the attributes and attribute values included in the profile are indicative of a fraudulent end-user profile 144 or are indicative of a valid end-user profile 124. The method of generating queries for identifying fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 by a fraud detection system 104 commences at 402.

[0095] At 404, the fraud detection system 104 identifies a subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 from a larger set of end-user profiles that may contain both valid end-user profiles 124 and fraudulent end-user profiles 144. In some instances, the identification of the subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 is performed manually, for example by a customer service representative or other system user who manually identifies those end-user profiles that are either fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144. In other instances, the identification of the subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 is performed autonomously by the fraud detection system 104. In yet other instances, the fraud detection system 104 may identify those end-user profiles included in the subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 for verification by a customer service representative or other authorized system user prior to using the subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 in generating one or more queries.

[0096] At 406, the fraud detection system 104 provides the fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 included in the subset to one or more query generation sub-systems. Advantageously, as new fraudulent end-user profiles 144 are received and identified, such profiles are provided to the query generation sub-system for incorporation into algorithms capable of producing new or additional queries that are able to identify and address the threat posed by these new fraudulent end-user profiles 144. Such a system thereby provides ongoing identification and removal of fraudulent end-user profiles 144.

[0097] At 408, using the fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 included in the subset, the query generation sub-system determines fraudulent end-user attributes and attribute values and combinations of fraudulent end-user attributes and attribute values that appear with a frequency above a defined threshold in the number of the fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144 included in the subset. In at least some instances, the query generation sub-system may generate data indicative of such queries based on one or more passes through the subset of fraudulent end-user profiles 144 and/or suspected fraudulent end-user profiles 144. For example, the query generation sub-system may perform a first pass that identifies the attribute value “Richmond, VA” associated with the end-user attribute of “end-user location” as appearing with a frequency greater than a first defined threshold (e.g., greater than 30%, 40%, 50%, 60%, or 70% of the end-user profiles included in one or more clusters) in a number of end-user profiles included in a single cluster. In such an instance might contain a pseudocode query such as “end-user location=Richmond, VA” to identify end-user profiles that include an attribute value of “Richmond, VA” associated with the end-user attribute “end-user location.”

[0098] At 410, the queries generated by the query generation sub-system are uploaded to the production fraud detection system 104. At 412, as new end-user profiles are received by the match service entity 102, the fraud detection system 104 can run one or more queries against new end-user profiles as they are received by the match service entity 102. In some instances, the fraud detection system 104 may generate a score or other valuation that provides an indication of whether the respective end-user profile represents a fraudulent end-user profile 144, a valid end-user profile 124, or an end-user profile requiring additional investigation prior to making a validity determination. In other instances, the fraud detection system 104 can autonomously prevent the addition of fraudulent end-user information to the end-user data store or database 110. The method of generating queries for identifying fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 by a fraud detection system 104 concludes at 414.

[0099] FIG. 5 shows a high-level method 500 of aggregating two or more queries for identifying fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 by a fraud detection system 104 such as that described in detail with regard to FIGS. 1-3, according to one illustrated embodiment. In at least some implementations, the fraud detection system 104 may combine or otherwise aggregate two or more queries into a single query to determine at signup whether a new end-user profile is a suspected fraudulent end-user profile 144. By combining or aggregating two or more queries into a single query, the fraud detection system is advantageously able to more accurately distinguish valid end-user profiles 124 from fraudulent end-user profiles 144. The low-level method 500 of running a combined or aggregate query that includes two or more queries against an end-user profile to determine whether the end-user profile is a fraudulent end-user profile 144 or a suspected fraudulent end-user profile 144 may be employed as part of, or in addition to, the high level method 400 (FIG. 4). The method 500 of generating two or more combined or aggregated queries for identifying fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 by a fraud detection system 104 concludes at 514.

files **144** and suspected fraudulent end-user profiles **144** by a fraud detection system **104** commences at **502**.

[0100] At **504**, the query generation sub-system generates one or more additional queries using subsequent passes through the subset of fraudulent end-user profiles **144** and/or suspected fraudulent end-user profiles **144**. In one instance, the query generation sub-system may generate an additional query based on another end-user attribute value that exceeds a second defined threshold (e.g., greater than 30%, 40%, 50%, 60%, or 70% of the end-user profiles included in one or more clusters). Continuing with the example in FIG. 4, the query generation sub-system may identify the attribute value “female” associated with the attribute “end-user gender” as appearing in fraudulent or suspected fraudulent end-user profiles **144** with a frequency greater than a second defined threshold (e.g., greater than 30%, 40%, 50%, 60%, or 70% of the end-user profiles included in one or more clusters). In such an instance, the query generation sub-system may provide a combined or compound pseudocode query such as, “end-user location =Richmond, VA AND end-user gender =female” to identify end-user profiles having an attribute value of “Richmond, VA” associated with the end-user attribute “end-user location” and an attribute value of “female” associated with the end-user attribute “end-user gender.”

[0101] In another example, again continuing with the example in FIG. 4, the query generation sub-system may, on occasion, perform a second pass specifically directed to the end-user attribute “end-user location” that identifies the attribute value “Petersburg, VA” as appearing in fraudulent or suspected fraudulent end-user profiles **144** with a frequency greater than a second defined threshold (e.g., greater than 30%, 40%, 50%, 60%, or 70% of the end-user profiles included in one or more clusters). In such an instance, the query generation sub-system may provide a combined or compound pseudocode query such as, “end-user location =Richmond, VA OR Petersburg, VA” to identify end-user profiles having the attribute value of “Richmond, VA” or “Petersburg, VA” associated with the end-user attribute “end-user location.” The method of generating two or more combined or aggregated queries for identifying fraudulent end-user profiles **144** and suspected fraudulent end-user profiles **144** by a fraud detection system **104** concludes at **506**.

[0102] FIG. 6 shows a low-level method **600** of generating running by a fraud detection system **104** one or more queries against the one or more end-user data stores or databases **110** to identify fraudulent end-user profiles **144** and suspected fraudulent end-user profiles **144** in the end-user data store or database **110**, according to one illustrated embodiment.

[0103] In at least some implementations, the fraud detection system **104** run one or more queries provided by the query generation sub-system against all or a portion of the valid end-user profiles **124** and/or fraudulent end-user profiles **144** stored, retained, or otherwise maintained in the end-user data store or database **110**. Running queries against the valid end-user profiles **124** and/or fraudulent end-user profiles **144** included in the data store or database **110** advantageously provides for the ongoing detection and removal from the data store or database **110** of those end-user profiles that include attributes and attribute values and/or combinations of attributes and attribute values indicative of fraudulent end-user profiles **144** and suspected fraudulent end-user profiles **144**. The low-level method **600**

of running one or more queries against some or all of the end-user profiles included in one or more end-user data stores or databases **110** may be employed as part of, or in addition to, the high level method **400** (FIG. 4). The method **600** of running one or more queries against the one or more end-user data stores or databases **110** to identify fraudulent end-user profiles **144** and suspected fraudulent end-user profiles **144** included in the data store or database commences at **602**.

[0104] At **604**, the fraud detection system **104** runs one or more queries generated by the query generation sub-system against some or all of the valid end-user profiles **124** and/or fraudulent end-user profiles **144** that are stored, retained, or otherwise maintained in the one or more end-user profile data stores or databases **110**. In some instances, queries may be run against some or all of the valid end-user profiles **124** and fraudulent end-user profiles **144** that are stored, retained, or otherwise maintained in the one or more end-user profile data stores or databases **110** on a defined schedule or at defined intervals. In some instances, queries may be run against some or all of the valid end-user profiles **124** and/or fraudulent end-user profiles **144** that are stored, retained, or otherwise maintained in the one or more end-user profile data stores or databases **110** at discretionary intervals selected by one or more matching service entity **102** system users and/or administrators. The method of running one or more queries against the one or more end-user data stores or databases **110** to identify fraudulent end-user profiles **144** and suspected fraudulent end-user profiles **144** included in the data store or database concludes at **606**.

[0105] FIG. 7 shows a low-level method **700** of determining the probability that an end-user profile included in a set of end-user profiles is either of a fraudulent end-user profile **144** or suspected fraudulent end-user profile **144**, according to one illustrated embodiment. The low level method **700** of determining the probability that an end-user profile is a fraudulent end-user profile **144** or a suspected fraudulent end-user profile **144** may be employed as part of, or in addition to, the high level method **400** (FIG. 4). The method **700** of determining the probability that an end-user profile included in a set of end-user profiles is either of a fraudulent end-user profile **144** or suspected fraudulent end-user profile **144** commences at **702**.

[0106] At **704**, the fraud detection system **104** computes values indicative of the initial probability ($p(V)$) that an end-user profile is a valid end-user profile **124** or the initial probability ($p(S)$) that an end-user profile is a fraudulent end-user profile **144**. In some instances, the probabilities $p(V)$ and $p(S)$ are determined based on a population of end-user profiles accumulated over a defined time period (last 3 months, last 6 months, last 12 months, etc.). In other instances, the probabilities $p(V)$ and $p(S)$ are determined based on a fixed population of end-user profiles (100 end-user profiles, 1000 end-user profiles, 10,000 end-user profiles, etc.) selected either at random or using a defined selection process. Within the selected population, the probability that a given end-user profile is valid is given by:

$$p(V) = \frac{(\text{number of valid end user profiles in population})}{(\text{total number of end user profiles in population})} \quad (1)$$

[0107] Within the selected population, the probability that a given end-user profile is fraudulent is given by:

$$p(S) = \frac{(\text{number of fraudulent end user profiles in population})}{(\text{total number of end user profiles in population})} \quad (2)$$

[0108] At 706, for each attribute value or combination of attribute values (x_i) included in each valid end-user profile 124 and in each fraudulent end-user profile 144, the fraud detection system 104 determines the likelihood that the particular attribute value or combination of attribute values will appear. Within the selected population, the likelihood that an attribute value or combination of attribute values (x_i) is included appears in a valid end-user profile 124 is given by:

$$p(x_i | V) = \frac{(\# \text{ of valid end user profiles with attribute } x_i)}{(\text{total number of valid end user profiles in population})} \quad (3)$$

[0109] Within the selected population, the likelihood that an attribute value or combination of attribute values (x_i) is included or appears in a fraudulent end-user profile 144 is given by:

$$p(x_i | S) = \frac{(\# \text{ of fraudulent end user profiles with attribute } x_i)}{(\text{total number of fraudulent end user profiles in population})} \quad (4)$$

[0110] At 708, the fraud detection system determines a value indicative of a conditional probability that a respective end-user profile is fraudulent using:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)} \quad (5)$$

[0111] Where $\{x_i\}$ includes all possible attribute values and M represents the total number of possible attribute values for the attributes logically associated with an end-user profile in the matching service entity 102. Where $\{x_i\}$ includes only single attribute values, the conditional probability determined by equation (5) represents a naïve Bayesian system. The conditional probability generated by the fraud detection system 104 can provide a score or valuation of a particular end-user profile, thereby providing an indication of the likelihood that the respective user profile is fraudulent.

[0112] Advantageously, the determination of a value indicative of a conditional probability that a respective end-user profile is fraudulent as determined using equation (5) above may be performed repeatedly across some or all of the population of end-user attribute value data stored, retained, or otherwise maintained in the data store or database 110. Such determination of a value indicative of a conditional probability that a respective end-user profile is fraudulent may be initiated at regular or defined intervals (every 4 hours, daily, weekly, etc.), at uneven intervals, and/or manually by a matching service entity 102 adminis-

trator. By repeating the determination of a value indicative of a conditional probability that a respective end-user profile is fraudulent, the fraud detection system is able to quickly respond to changes in the attribute values appearing in fraudulent end-user profiles 144. For example, if a fraudulent end-user 130 has recently generated a number of fraudulent end-user profiles 144 on a processor-based device 132 using the Linux operating system, the conditional probability value that a given end-user profile is fraudulent given by equation (5) will increase for those end-users connected to the matching service entity 102 via a processor-based device 122, 132 using the Linux operating system. The conditional probability determined using equation (5) is therefore beneficially indicative of changes in both valid end-user and fraudulent end-user behavior on a real-time or near real-time basis.

[0113] In at least some instances, a single attribute value may be logically associated with only valid end-users 120 or fraudulent end-users 130. In such instances, a naïve Bayesian system encountering a profile containing the single attribute value or combination of attribute values will always be identified as either valid or fraudulent, creating a tautology. To avoid this result, where an attribute value is found logically associated with only valid end-users 120 or fraudulent end-users 130, a fractional end-user value is added to the attribute value, thereby avoiding the 100% valid or 100% fraudulent conditional probability determination and permitting the fraud detection system 104 to consider these attribute values in future conditional probability determination cycles. The value of the fractional end-user added to the population in such instances is determined on an attribute-by-attribute basis.

[0114] At 710, the fraud detection system 104 ranks each end-user profile into one of a number of groups based at least in part on the conditional probability score or valuation associated with the end-user profile. In some instances, the end-user profiles are ranked upon creation and prior to addition to the end-user data store or database 110. In some instances, all or a portion of the end-user profiles in the end-user data store or database 110 are ranked using the most recently determined conditional probability information.

[0115] The end-user profiles may be ranked into any number of groups based on the determined conditional probability that the end-user profile is fraudulent. For example, in one implementation each end-user profile may be classified into one of the following groups based in whole or in part on the determined conditional probability that the respective end-user profile is fraudulent: a clearly fraudulent end-user profile that is automatically deleted from the end-user database 110 or automatically not added to the end-user database 110 by the fraud detection system 104; a suspicious or potentially fraudulent end-user profile that is forwarded by the fraud detection system 104 for further manual review and/or confirmation prior to deletion from the end-user database 110 (existing end-user profiles) or addition to the end-user database 110 (new end-user profiles); and, a likely valid end-user profile that is retained in the end-user database 110 (existing end-user profiles) or added to the end-user database 110 (new end-user profiles). At 712, the fraud detection system 104 autonomously deletes end-user profiles logically associated with identified fraudulent end-users 130.

[0116] At 714, the fraud detection system 104 autonomously transfers data indicative of those end-user profiles identified as suspected of being fraudulent end-user profiles 144 at 710 to a customer service representative for manual review and confirmation. In some instances, the customer service representative can manually delete suspected fraudulent end-user profiles 144 after confirmation that the end-user profile is fraudulent. In some instances, the customer service representative can manually clear suspected fraudulent end-user profiles 144 after confirmation that the end-user profile is valid. In some instances, the customer service representative can forward suspected fraudulent end-user profiles 144 for additional review. The method 700 of determining the conditional probability that an end-user profile included in a set of end-user profiles is a fraudulent end-user profile 144 or suspected fraudulent end-user profile 144 concludes at 716.

[0117] FIG. 8 shows a low-level method 800 of determining the conditional probability that an end-user profile is either of a fraudulent end-user profile 144 or suspected fraudulent end-user profile 144 using a combination of attribute values in a semi-naïve Bayesian system, according to one illustrated embodiment. The low level method 800 of determining the probability that an end-user profile is a fraudulent end-user profile 144 or a suspected fraudulent end-user profile 144 using a combination of attribute values in a semi-naïve Bayesian system may be employed as part of, or in addition to, the low level method 700 (discussed in detail with regard to FIG. 7). The method 800 of determining the probability that an end-user profile is a fraudulent end-user profile 144 or a suspected fraudulent end-user profile 144 using a combination of attribute values in a semi-naïve Bayesian system commences at 802.

[0118] At 804, the fraud detection system determines a value indicative of a conditional probability that a respective end-user profile is fraudulent using:

$$p(S | \{x_i\}) = \frac{p(S) \prod_{i=1}^M p(x_i | S)}{p(S) \prod_{i=1}^M p(x_i | S) + p(V) \prod_{i=1}^M p(x_i | V)} \quad (6)$$

[0119] Where $\{x_i\}$ includes all possible attribute values and/or combinations of attribute values and M represents the total number of possible attribute values and/or combination of attribute values for the attributes logically associated with an end-user profile in the matching service entity 102. Where $\{x_i\}$ represents both attribute values and combinations of attribute values, the conditional probability determined by equation (6) represents a semi-naïve Bayesian system. Generally, a greater level of confidence is placed in the conditional probability determined for a semi-naïve Bayesian system over a naïve Bayesian system. The conditional probability generated by the fraud detection system 104 can provide a score or valuation of a particular end-user profile, thereby providing an indication of the likelihood that the respective user profile is fraudulent.

[0120] Surprisingly and advantageously, it has been found that fraudulent end-user profiles 124 typically contain one of a number of defined or known combinations of attribute values, thereby permitting the determination of a conditional probability value using a semi-naïve Bayesian system. In at least some instances, the known combinations of attribute values may be attributable at least in part to use of repeated

instances of the software, scripts, or machine-readable instructions executing on various processor-based devices as bots used to generate the fraudulent user profiles 144. Since software, scripts, or machine-readable instructions draw from a limited number of attribute values (e.g., based on preferred virtual end-user 140 characteristics specified or defined by the fraudulent end-user 130), it has been observed that specific combinations of attribute values appear in greater frequency in clusters of software, script, or machine-readable instruction generated fraudulent end-user profiles 144 than in clusters containing a greater proportion of valid end-user profiles 124. The method 800 of determining the probability that an end-user profile is a fraudulent end-user profile 144 or a suspected fraudulent end-user profile 144 using a combination of attribute values in a semi-naïve Bayesian system concludes at 806.

[0121] FIG. 9 shows a low-level method 900 of communicating end-user attribute and attribute value data logically associated with determined fraudulent end-user profiles 144 and/or determined suspected fraudulent end-user profiles 144 detected using the naïve or semi-naïve Bayesian systems of methods 700 or 800 as training examples for query generation, according to one illustrated embodiment. The low level method 900 of communicating determined fraudulent end-user profiles 144 and/or determined suspected fraudulent end-user profiles 144 detected using the naïve or semi-naïve Bayesian systems of methods 700 or 800 as training examples for query generation may be employed as part of, or in addition to, the high level method 400 (FIG. 4). The method 900 of communicating determined fraudulent end-user profiles 144 and/or determined suspected fraudulent end-user profiles 144 detected using the naïve or semi-naïve Bayesian systems of methods 700 or 800 as training examples for query generation commences at 902.

[0122] At 904, the fraud detection system 104 communicates end-user attribute and attribute value data to the query generation sub-system for use as training examples for query generation. By providing training examples of known fraudulent end-user profiles 144 and suspected fraudulent end-user profiles 144 to the query generation sub-system, the accuracy of the queries produced by the query generation sub-system is advantageously improved.

[0123] Advantageously, since the determination of a value indicative of a conditional probability that a respective end-user profile is fraudulent as determined using equations (5) and (6) above may be performed repeatedly and is quickly updated based on end-user attribute and attribute value data supplied to the matching service entity 102, the query generation sub-system receives end-user attribute and attribute value data logically associated with and indicative of known and suspected fraudulent end-user profiles 144 on a real-time or near-real time basis. Such permits the query generation sub-system to produce queries that are able to accurately identify fraudulent or suspected fraudulent end-user profiles 144 and are also able to advantageously adapt to changes in attribute values or combinations of attribute values logically associated with fraudulent and suspected fraudulent end-user profiles 144. The method 900 of communicating determined fraudulent end-user profiles 144 and/or determined suspected fraudulent end-user profiles 144 detected using the naïve or semi-naïve Bayesian systems of methods 700 or 800 as training examples for query generation concludes at 906.

[0124] FIG. 10 shows a low-level method 1000 of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles 144 by the fraud detection system 104, according to one illustrated embodiment. The low level query generation method 1000 may be employed as part of, or in addition to, the high level method 400 (FIG. 4). The method 1000 of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles 144 by the fraud detection system 104 commences at 1002.

[0125] At 1004, the fraud detection system 104 computes data indicative of at least one representation of dissimilarity existent in a set of end-user profiles that includes any number of valid end-user profiles 124 and any number of fraudulent end-user profiles 144.

[0126] At 1006, the fraud detection system 104 clusters end-user profiles based on the at least one representation of dissimilarity determined at 1004. In at least some implementations, such clustering is performed by applying one or more clustering algorithms to a set of end-user profiles that may include new end-user profiles, existing end-user profiles, or a combination thereof to provide a set of clusters, each including at least one end-user profile. In at least some implementations, the clustering algorithm may include one or more agglomerative clustering algorithms. In at least some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on a determined representation of dissimilarity in the set of end-user profiles 124, 144. In at least some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on minimizing an average of dissimilarities between one determined user-profile cluster and another determined user profile cluster. In at least some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on clustering end-user profiles based at least in part on the dissimilarity matrices of the respective end-user profiles 124, 144. In some instances, the clustering algorithm performed by the processor(s) 212 may include an agglomerative nesting clustering algorithm that clusters end-user profiles 124, 144 using at least one of: a nearest neighbor method, a furthest neighbor method, a pair group average method, a centroid method, a median method, and Ward's method. In some instances, the clustering algorithm performed by the processor(s) 212 may include a divisive analysis clustering of end-user profiles 124, 144. In some instances, the clustering algorithm performed by the processor(s) 212 may include normalizing the values associated with one or more end-user attributes 126, 146 prior to performing the cluster analysis.

[0127] At 1008, the fraud detection system 104 selects one or more end-user profile clusters for additional investigation. Any defined cluster selection criterion, process, or algorithm may be applied by the fraud detection system 104 to select the one or more end-user profile clusters. In at least some implementations, the clustering performed by the fraud detection system 104 at 1006 generates a dendrogram such as the illustrative dendrogram 300 depicted in FIG. 3.

[0128] The fraud detection system 104 autonomously or semi-autonomously selects various clusters resulting from the clustering algorithm that are suspected of including a number of fraudulent end-user profiles 144. For example, again referring to the dendrogram 300 depicted in FIG. 3, in at least some implementations, the processor(s) 212 can

select a horizontal "cut" across the dendrogram 300. In some instances, some or all of the end-user profile clusters falling below the "cut" line on the dendrogram are selected for further investigation. Cluster generation and selection algorithms other than the dendrogram discussed in detail above may be substituted.

[0129] At 1010, the fraud detection system 104 makes a first pass through each end-user profile cluster selected at 1008 to identify respective end-user attributes and attribute values and/or combinations of end-user attributes and attribute values appearing within the respective end-user profile cluster at a frequency exceeding a first defined threshold. In recognition of the similarity existed between processor generated fraudulent end-user profiles 144, end-user attributes and attribute values and/or combinations of end-user attributes and attribute values appearing within the respective end-user profile cluster at a frequency above a first defined threshold (e.g., greater than "x" percent of the population of the respective cluster) are flagged or otherwise identified as indicative of the existence of a fraudulent end-user profile 144.

[0130] The fraud detection system 104 may employ one or more algorithms to perform, at least in part, the depth-first search for end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that occur with a frequency exceeding the first defined threshold (i.e., the "support threshold") in the subset of end-user profiles included in each of the clusters selected at 1008. The fraud detection system 104 may perform such a depth-first search for frequent end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that occur with a frequency exceeding the first defined threshold can include at least one of: an Eclat algorithm, an Apriori algorithm, an FP-growth algorithm, a GUHA Procedure ASSOC, or an OPUS search.

[0131] By way of illustrative example, for the end-user attribute "end-user processor-based device operating system" a cluster including 100 end-user profiles may have the following attribute value distribution: OS/X Version 8=62%; OS/X Version 7.5=33%; other Unix based O/S=5%. If the support threshold were set at 60% (i.e., an end-user attribute value appearing in more than 60% of the end-user profiles included in the respective cluster) the processor would flag the end-user attribute of "end-user processor-based device operating system" and the end-user attribute value of "OS/X Version 8" as indicative of a suspected fraudulent end-user profile 144.

[0132] At 1012, using the attributes or combinations of attributes identified as exceeding the first threshold (i.e., the "support threshold") at 1010, the fraud detection system 104 makes a second pass through the cluster identified as including the attributes or combinations of attributes identified as exceeding the first threshold. During the second pass, the fraud detection system 104 identifies additional end-user attributes and attribute values and/or combinations of end-user attributes and attribute values occurring with a frequency above a second defined threshold in the respective end-user profile cluster. In at least some implementations, end-user attributes and attribute values and/or combinations of end-user attributes and attribute values occurring within the respective end-user cluster at a frequency above a second defined threshold (e.g., greater than "y" percent of the

population of the respective cluster) are flagged or otherwise identified as indicative of the potential existence of a fraudulent end-user profile 144.

[0133] In at least some implementations, the fraud detection system 104 uses an algorithm to perform, at least in part, the search for end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that occur with a frequency exceeding the second defined threshold (i.e., the “combine threshold”) in the subset of end-user profiles included in each of the clusters selected at 1008.

[0134] Again by way of illustrative example as using the same data as above, the end-user attribute “end-user processor-based device operating system” was identified in the first pass as an attribute relevant to identifying fraudulent end-users 140. Specifically, an attribute value of “OS/X Version 8” was identified as indicative of a potentially fraudulent end-user profile 144. Assuming a second threshold of 30% (i.e., a “combine threshold” equal to 30%), in the second pass, the fraud detection system 104 would further identify the attribute value “OS/X Version 7.5” (included in 33% of the end-user profiles included in the respective end-user profile cluster) as indicative of a suspected fraudulent end-user profile 144. Thus, after completing the second pass, the fraud detection system 104 would identify end-user profiles that include an end-user attribute value of either “OS/X Version 8” or “OS/X Version 7.5” associated with the end-user attribute “end-user processor-based device operating system” as indicative of a suspected fraudulent end-user profile 144.

[0135] At 1014, the fraud detection system 104 prepares one or more simple or compound queries to identify known or suspected fraudulent end-user profiles 144 based at least in part on the end-user attribute and attribute values and/or combinations of end-user attributes and attribute values identified in the first pass analysis at 1010. Optionally, the fraud detection system 104 prepares one or more simple or compound queries to identify known or suspected fraudulent end-user profiles 144 based at least in part on the end-user attribute and attribute values and/or combinations of end-user attributes and attribute values identified in the second pass analysis at 1012.

[0136] Continuing with the illustrative example developed in detail above, a first illustrative query to detect a suspected fraudulent end-user profile 144 may include querying the end-user profile using the pseudocode query “end-user processor-based device operating system=OS/X Version 8 or OS/X Version 7.5.” An affirmative response to the query may indicate the end-user profile being queried is a fraudulent end-user profile 144.

[0137] In at least some instances, the fraud detection system 104 may identify a second end-user attribute and attribute value as exceeding the defined first-pass threshold value. In such an instance, the fraud detection system 104 may generate a compound query (i.e., a query composed of two or more individual attributes and respective attribute values) to detect whether an end-user profile is a suspected fraudulent end-user profile 144.

[0138] Continuing with the illustrative example developed in detail above, during the first pass, the fraud detection system may also identify the attribute value “Richmond, VA” associated with the end-user attribute of “end-user location” as appearing with a frequency greater than a first defined threshold. In such an instance, the fraud detection system may combine the attribute and attribute values

identified in the first and second passes to provide a compound query. An illustrative compound query to detect a suspected fraudulent end-user profile 144 may include querying the end-user profile using the pseudocode query “end-user processor-based device operating system=OS/X Version 8 OR OS/X Version 7.5 AND end-user location=Richmond, VA.” An affirmative response to the query may indicate the end-user profile being queried is a fraudulent end-user profile 144. The method 1000 of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles 144 by the fraud detection system 104 concludes at 1016.

[0139] FIG. 11 shows a low-level method 1100 of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles 144 by the fraud detection system 104, according to one illustrated embodiment. The low level query generation method 1100 may be employed as part of, or in addition to, the high level method 400 (FIG. 4). The method 1100 of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles 144 by the fraud detection system 104 commences at 1102.

[0140] At 1104, the fraud detection system 104 computes a dissimilarity matrix reflecting the degree of dissimilarity existent in the attributes and attribute values and/or combinations of end-user attributes and attribute values for each end-user profile included in a set of end-user profiles. The set of end-user profiles can be of any size and can contain data logically associated with any number of fraudulent end-users 130.

[0141] At 1106, the fraud detection system 104 normalizes the attribute values for each of the attributes included in the end-user profiles.

[0142] At 1108, the fraud detection system 104 applies at least one clustering algorithm to cluster end-user profiles in a manner that minimizes the average of dissimilarities between the end-user profiles included in each respective cluster. In at least some implementations, the fraud detection system uses the dissimilarity matrices generated at 1104 and the normalized attribute values determined at 1106 to perform at least a portion of the end-user profile clustering. In at least some implementations, the clustering algorithm may include one or more agglomerative clustering algorithms. In some implementations, the clustering algorithm may include an agglomerative clustering algorithm that is based at least in part on clustering end-user profiles based at least in part on the dissimilarity matrices of the respective end-user profiles generated at 1104.

[0143] At 1110, the fraud detection system 104 selects one or more end-user profile clusters for additional investigation using a dendrogram such as the illustrative dendrogram 300 depicted in FIG. 3 generated based on the clustering algorithm applied to the end-user profiles at 1108. The fraud detection system 104 autonomously or semi-autonomously selects various end-user profile clusters falling beneath a defined “cut” line on the dendrogram are selected for further investigation.

[0144] At 1112, the fraud detection system 104 makes a first pass through each end-user profile cluster selected at 1008 to identify respective end-user attributes and attribute values and/or combinations of end-user attributes and attribute values appearing within the respective end-user profile cluster at a frequency exceeding a first defined threshold. In recognition of the similarity existed between processor

generated fraudulent end-user profiles **144**, end-user attributes and attribute values and/or combinations of end-user attributes and attribute values appearing within the respective end-user profile cluster at a frequency above a first defined threshold are flagged or otherwise identified as indicative of the existence of a fraudulent end-user profile **144**.

[0145] The fraud detection system **104** may employ one or more algorithms to perform, at least in part, the depth-first search for end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that occur with a frequency exceeding the first defined threshold (i.e., the “support threshold”) in the subset of end-user profiles included in each of the clusters selected at **1008**. The fraud detection system **104** may perform such a depth-first search for frequent end-user attributes and attribute values and/or combinations of end-user attributes and attribute values that occur with a frequency exceeding the first defined threshold can include at least one of: an Eclat algorithm, an Apriori algorithm, an FP-growth algorithm, a GUHA Procedure ASSOC, or an OPUS search.

[0146] At **1114**, using the attributes or combinations of attributes identified as exceeding the first threshold (i.e., the “support threshold”) at **1112**, the fraud detection system **104** makes a second pass through the cluster identified as including the attributes or combinations of attributes identified as exceeding the first threshold. During the second pass, the fraud detection system **104** identifies additional end-user attribute values occurring with a frequency above a second defined threshold in the respective end-user profile cluster. In at least some implementations, end-user attributes and attribute values and/or combinations of end-user attributes and attribute values occurring within the respective end-user cluster at a frequency above a second defined threshold are flagged or otherwise identified as indicative of the potential existence of a fraudulent end-user profile **144**. The method **1100** of generating one or more queries useful for identifying fraudulent and suspected fraudulent end-user profiles **144** by the fraud detection system **104** concludes at **1102**.

[0147] The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, schematics, and examples. Insofar as such block diagrams, schematics, and examples contain one or more functions and/or operations, it will be understood by those skilled in the art that each function and/or operation within such block diagrams, flowcharts, or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. In one embodiment, the present subject matter may be implemented via Application Specific Integrated Circuits (ASICs). However, those skilled in the art will recognize that the embodiments disclosed herein, in whole or in part, can be equivalently implemented in standard integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more controllers (e.g., microcontrollers) as one or more programs running on one or more processors (e.g., microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of ordinary skill in the art in light of this disclosure.

[0148] Those of skill in the art will recognize that many of the methods or algorithms set out herein may employ additional acts, may omit some acts, and/or may execute acts in a different order than specified.

[0149] In addition, those skilled in the art will appreciate that the mechanisms taught herein are capable of being distributed as a program product in a variety of forms, and that an illustrative embodiment applies equally regardless of the particular type of signal bearing media used to actually carry out the distribution. Examples of signal bearing media include, but are not limited to, the following: recordable type media such as floppy disks, hard disk drives, CD ROMs, digital tape, and computer memory.

[0150] The various embodiments described above can be combined to provide further embodiments. All of the commonly assigned US patent application publications, US patent applications, foreign patents, and foreign patent applications referred to in this specification and/or listed in the Application Data Sheet, including but not limited to U.S. Provisional Application No. 61/911,908, filed Dec. 4, 2013, U.S. application Ser. Nos. 14/561,004, filed Dec. 4, 2014, and 15/782,576, filed Oct. 12, 2017, are incorporated herein by reference, in their entirety. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

[0151] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

1. A method of detecting suspected fraudulently generated user identity profiles in an online network environment using a fraud detection system that includes at least one processor and at least one non-transitory processor-readable medium that stores processor-executable instructions, the online network environment including an attribute value for an attribute associated with the user identity profiles, and including an additional attribute value that occurs frequently in the set of profiles, the method comprising:

determining, via the at least one processor, whether the attribute value is linked to a high conditional probability based on the attribute value that a profile is fraudulently generated;

identifying, via the at least one processor, a set of profiles likely to be fraudulently generated based on the attribute value and the high conditional probability;

querying, using the fraud detection system having at least one processor and at least one non-transitory processor-readable medium that stores processor-executable instructions, a data store containing profile information to identify profiles suspected of being fraudulently generated based at least in part on the attribute value and the additional attribute value; and

quarantining the identified profiles.

2. The method of claim 1, wherein determining whether the attribute value is linked to a high conditional probability of being a fraudulent profile includes computing the initial probabilities of a profile being a fraudulent profile [p(S)] or a valid profile [p(V)].

3. The method of claim 1, wherein determining whether the attribute value is linked to a high conditional probability includes computing the value indicative of a conditional probability that the respective profile is fraudulent according to:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)}.$$

4. The method of claim 1, wherein the attribute value includes one or more of:

- a Hyper Text Transfer Protocol (http) referrer associated with the respective profile;
- an Internet Protocol (IP) of signup and last logins associated with the respective profile;
- one or more cookies used to track individual computers associated with the respective profile;
- one or more cookies that contain a user identifier of the most recent users to log in on using a given instance of a processor-based device; and
- IP blocks of signup and at least two most recent logins associated with the respective profile.

5. The method of claim 1, further comprising: screening the identified profiles suspected of being fraudulently generated.

6. The method of claim 1, further comprising: providing the designated profiles to a front end system for possible deletion.

7. A user identity profile detection system in an online network environment to detect at least one of accounts or related profiles suspected of being undesirable, the online network environment including an initial probability value based at least in part on historical profile data, and including an additional attribute value that occurs frequently in the set of profiles, the system comprising:

- at least one processor; and
- at least one non-transitory processor-readable medium that stores processor-executable instructions, wherein the at least one processor:
 - determine whether the initial probability value is linked to a value indicative of a high conditional probability based on the initial probability value that a profile is undesirable;
 - identify a set of profiles likely to be undesirable based on the initial probability value and the high conditional probability;
 - query a data store containing profile information to identify profiles suspected of being undesirable based at least in part on the initial probability value and the additional attribute value; and
 - quarantine the identified profiles.

8. The system of claim 7, wherein the at least one processor computes a value indicative of the initial probabilities of a profiles being at least one of an undesirable-profile [p(S)] or a valid profile [p(V)].

9. The system of claim 7, wherein the at least one processor computes a value indicative of a conditional

probability that the respective profile is undesirable includes computing the value indicative of a conditional probability that the respective profile is undesirable according to:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)}.$$

10. The system of claim 7, wherein the at least one processor ranks the profiles into groups using the high conditional probability based on the initial probability value that the profile is undesirable.

11. The system of claim 7, wherein the attribute value includes one or more of:

- a Hyper Text Transfer Protocol (http) referrer associated with the respective profile;
- an Internet Protocol (IP) of signup and last logins associated with the respective profile;
- one or more cookies used to track individual computers associated with the respective profile;
- one or more cookies that contain a user identifier of the most recent users to log in on using a given instance of a processor-based device; and
- IP blocks of signup and at least two most recent logins associated with the respective profile.

12. The system of claim 7, wherein the at least one processor generates at least one output logically associated with each profile, the at least one output indicative of at least one of the following: a deletion indicator, a clearance indicator, or a further investigation indicator.

13. A fraudulent user identity profile detection system in an online network environment to detect at least one of accounts or related profiles suspected of being fraudulently generated, the online network environment including an attribute value for an attribute associated with the user identity profiles, and including an additional attribute value that occurs frequently in the set of profiles, the system comprising:

- at least one processor; and
- at least one nontransitory processor-readable medium that stores processor-executable instructions, wherein the at least one processor:
 - determine whether the attribute value is linked to a high conditional probability based on the attribute value that a profile is fraudulently generated;
 - identify a set of profiles likely to be fraudulently generated based on the attribute value and the conditional probability;
 - query a data store containing profile information to identify profiles suspected of being fraudulently generated based at least in part on the attribute value and the additional attribute value; and
 - quarantine the identified profiles.

14. The system of claim 13, wherein the at least one processor computes a value indicative of an initial probabilities of a profiles being at least one of a fraudulent profile [p(S)] or a valid profile [p(V)].

15. The system of claim 13, wherein the at least one processor computes a value indicative of a conditional probability that the respective profile is fraudulent includes computing the value indicative of a conditional probability that the respective profile is fraudulent according to:

$$p(S | \{x_i\}) = \frac{p(S) \prod_i^M p(x_i | S)}{p(S) \prod_i^M p(x_i | S) + p(V) \prod_i^M p(x_i | V)}.$$

16. The system of claim **13**, wherein the at least one processor ranks the profiles into groups using the high conditional probability based on the initial probability value that the profile is fraudulent.

17. The system of claim **13**, wherein the attribute value includes one or more of:

a Hyper Text Transfer Protocol (http) referrer associated with the respective profile;

an Internet Protocol (IP) of signup and last logins associated with the respective profile;

one or more cookies used to track individual computers associated with the respective profile;

one or more cookies that contain a user identifier of the most recent users to log in on using a given instance of a processor-based device; and

IP blocks of signup and at least two most recent logins associated with the respective profile.

18. The system of claim **13**, wherein the at least one processor generates at least one output logically associated with each profile, the at least one output indicative of at least one of the following: a deletion indicator, a clearance indicator, or a further investigation indicator.

19. The system of claim **13**, wherein the at least one processor screens the identified profiles suspected of being fraudulently generated.

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